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Feature-Driven Portfolio Architecture:

A Two-Layer Quantitative Asset Rotation Framework

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Abstract

We present a systematic multi-asset rotation framework built around a two-layer selection pipeline and a rigorous pre-modeling feature validation protocol, evaluated out-of-sample on 21 liquid ETFs over 2009–2026 (893 weekly rebalancing periods).

The central methodological contribution is a feature diagnostic conducted prior to any architectural decision, comprising cross-sectional and time-series Spearman Information Coefficient (IC) audits, a pairwise redundancy map, and an alpha-orthogonality test, applied to a candidate pool of fifteen signals spanning price, volume, and volatility dynamics. This diagnostic formally distinguishes features with cross-sectional predictive power (useful for ranking assets relative to one another) from features with time-series predictive power (useful for timing individual assets against their own history). This distinction has direct architectural consequences rarely made explicit in the quantitative literature, and is shown to hold even for features that are formally collinear, where standalone predictive strength and genuine incremental information diverge.

The selection pipeline encodes this distinction structurally. Layer 1 applies a cross-sectional momentum funnel (incorporating Point of Control crossovers and 52-week momentum; Jegadeesh and Titman, 1993; Moskowitz et al., 2012), narrowing the universe to trend-aligned candidates. Layer 2 then applies a time-series mean-reversion filter (utilizing Kalman filter deviations and sign-corrected moving-average slopes) to select the final portfolio holdings.

A portfolio construction ablation holds selection fixed and compares multiple weighting schemes in isolation. The optimal ScoreWeighted Dual-Layer Funnel achieves a pure out-of-sample Sharpe ratio of 1.38 (January 2022 – May 2026) with a statistically significant annualised alpha of 13.4% ($t = 2.87$, $p = 0.004$, HAC-robust), versus a benchmark Sharpe of 0.68 over the same window. Over the full evaluation period (2010/2011–2026) the strategy delivers a CAGR of 15.30% and a Sharpe ratio of 0.993 against SPY's 12.18% CAGR and 0.790 Sharpe, while significantly reducing maximum drawdown (-21.6% vs. -32.2%). Risk-parity methods (e.g., ERC, HRP; Maillard et al., 2010; López de Prado, 2016) underperform simpler allocation rules, attributable to their tendency to override strong directional conviction with variance geometry. The statistical robustness of these results is validated through Deflated Sharpe Ratio (DSR) and Probability of Backtest Overfitting (PBO) diagnostics (Bailey and López de Prado, 2014).

Finally, two machine-learning alternatives are evaluated against this deterministic architecture: a walk-forward XGBoost benchmark (Chen and Guestrin, 2016) on an expanded feature set, and a LightGBM (Ke et al., 2017) ensemble trained directly on a pairwise Learning-to-Rank (LambdaRank) objective (Burges et al., 2005; Burges, 2010b). Across all machine-learning variants, no data-driven model converts comparable predictive signal into an execution profile as operationally efficient as the two-layer funnel after accounting for transaction costs and weight-drift rebalancing friction. This confirms that in small-universe, weekly-frequency settings, an explicit, feature-validated structural prior outperforms highly flexible models that must discover the underlying market geometry unaided.

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1. Introduction

1.1 Motivation

Asset allocation and portfolio rotation are among the most actively studied problems in quantitative finance. The central question is deceptively simple: given a universe of investable assets, which subset should a portfolio hold this week, and in what proportions? Despite decades of academic research and the proliferation of machine learning methods in finance, this question remains open for multi-asset universes composed of heterogeneous instruments such as equity sector ETFs, fixed income, commodities, and real assets.

Two broad traditions have emerged to address it. The first, rooted in the cross-sectional momentum literature ((Jegadeesh and Titman, 1993; Asness et al., 2013), ranks assets relative to each other on momentum and related signals, selecting the top-performing subset. The second, rooted in time-series momentum (Moskowitz et al., 2012), evaluates each asset against its own history, taking long positions when the asset's own trend is positive and reducing or closing positions otherwise.

Both traditions have been extensively extended with machine learning. Gu et al. (2020) demonstrate that non-linear models trained on large panels of firm characteristics generate substantial cross-sectional alpha on equity universes of several hundred to several thousand stocks. Fischer and Krauss (2018) apply long short-term memory networks to S&P 500 constituent rotation. More recently, transformer architectures and graph neural networks have been applied to capture cross-asset dependencies (Chen et al., 2023).

Yet a systematic gap persists in the literature. The vast majority of machine learning applications in asset selection operate on large, homogeneous universes (typically 300 to 3,000 US equities with similar volatility profiles) denominated in the same currency, and subject to the same macroeconomic exposures. The implicit assumption is that the cross-section is large enough to provide the sample density required for non-linear models to generalise out of sample. When this assumption breaks down (as it does in multi-asset rotation across heterogeneous instruments) the performance of standard machine learning pipelines degrades sharply, often below simple momentum baselines.

This thesis investigates why this degradation occurs, identifies the statistical mechanism that drives it, and proposes an architecture that exploits the available signal more effectively on small heterogeneous universes.

1.2 The Core Problem: Feature Context Dependency

The central empirical finding of this thesis is that the same feature can exhibit a positive Information Coefficient (IC) in one statistical context and a negative or negligible IC in another. Specifically, we document that features computed on a cross-sectional panel (where assets are ranked relative to each other at the same point in time) systematically differ in predictive sign and magnitude from the same features computed in a time-series context, where each asset is evaluated against its own historical distribution.

This distinction is well-known in the practitioner literature (Grinold and Kahn, 2000) but rarely tested empirically on multi-asset universes, and never, to our knowledge, used as the primary architectural principle for portfolio construction. We show that assigning each feature to the layer in which its IC is statistically supported (cross-sectional features to an initial universe filter, time-series features to an asset-specific conviction signal) produces an out-of-sample Sharpe ratio of 0.997 versus 0.434 for a cross-sectional XGBoost baseline trained on the same feature set with the same walk-forward protocol.

1.3 Research Questions

This thesis addresses three interconnected questions:

RQ1. Do microstructure and momentum features derived from price, volume, and volatility exhibit statistically significant predictive power on a 21-asset multi-asset ETF universe, and does that predictive power depend on whether the feature is evaluated cross-sectionally or in time series?

RQ2. Does a machine learning model trained cross-sectionally on validated features add value over a structured rank-based filter on a small heterogeneous universe, and if not, what is the statistical explanation?

RQ3. What is the optimal construction architecture for a weekly rotational portfolio on a multi-asset ETF universe, and how sensitive is it to the choice of feature combination weights and position sizing scheme?

1.4 Contributions of this Work

This thesis makes three contributions to the quantitative finance literature:

1. Empirical documentation of cross-sectional vs time-series IC divergence on a multi-asset universe. We conduct a systematic IC audit on fifteen features across both statistical contexts, showing that features can have IC of opposite sign depending on the context of evaluation.

This finding motivates a two-layer architecture that respects the context in which each signal is statistically supported, rather than feeding all features indiscriminately into a single model.

2. A two-layer rank-based portfolio architecture grounded in IC evidence. We propose a pipeline in which a cross-sectional momentum funnel (Layer 2) pre-selects assets in trend relative to the universe, and a time-series conviction filter (Layer 1) identifies assets whose short-term deviation from their own Kalman fair value signals a favourable entry point. Feature combination within each layer follows the ICIR-weighted z-score sum of (Grinold and Kahn, 2000).

3. A systematic ablation study on construction choices. We isolate the contribution of each architectural component — layer topology, feature weights, and portfolio sizing scheme — through a walk-forward ablation, showing that the two-layer signal structure is the dominant source of alpha, while sizing scheme and feature weight precision are second-order effects.

1.5 Structure of the Document

The remainder of this thesis is organized as follows. Chapter 2 presents a comprehensive literature review that covers cross-sectional and time-series momentum, volatility risk controls, volume-based indicators such as the Point of Control, the Kalman filter as a fair-value estimator, portfolio construction methods, and machine learning paradigms in asset selection. Chapter 3 details the core empirical and quantitative methodology, beginning with Section 3.1.3 on data processing and descriptive statistics for the 21-ETF universe. Section 3.4 introduces the feature engineering and validation protocol, executing cross-sectional and time-series Spearman Information Coefficient audits, redundancy mapping, and alpha-orthogonality testing to resolve feature context dependency, which constitutes the central methodological contribution of this work. Section 3.10 establishes the two-layer rank architecture, evaluating its out-of-sample performance, weighting scheme ablations, turnover sensitivity, and statistical robustness. Section 3.11 evaluates an unstructured XGBoost cross-sectional benchmark to analyze its empirical performance and failure modes on heterogeneous ETF panels. Section 3.12 formulates a LightGBM LambdaRank framework, examining its mathematical mechanics, SHAP directional diagnostics, macro-regime interactions, and hysteresis dynamics against the deterministic baseline. Finally, Chapter 4 concludes with a synthesis of the main findings, an explicit discussion of methodological limitations, and actionable directions for future research.

2. Literature Review

2.1 Introduction

Financial prices are not one-dimensional. The observed price of an asset at any point in time is the outcome of a complex interaction between three distinct informational dimensions, each of which carries independent signals about future returns.

The first dimension is **price action**: the sequence of past prices encodes information about trend, momentum, and mean-reversion. The momentum anomaly (the tendency for recent winners to continue outperforming recent losers) is one of the most replicated findings in empirical finance Jegadeesh and Titman, 1993; Moskowitz et al., 2012, and market microstructure indicators such as the Kalman filter deviation capture short-term deviations from the asset's intrinsic trend.

The second dimension is **volume**: the distribution of trading activity across price levels reveals where market participants have collectively agreed on value. The Point of Control (the price level of maximum volume concentration within a rolling window) provides a volume-wise grounded reference level that is distinct from any price-return based signal. When price diverges from the volume-weighted centre of gravity, it carries information about both trend strength and potential reversion.

The third dimension is **volatility**: the level and regime of realized volatility encodes information about the risk environment. High volatility is associated with uncertainty, forced selling, and wider bid-ask spreads; low volatility is associated with trend persistence and favourable risk-adjusted entry conditions. Volatility signals are used in this thesis as a feature (short-term realized volatility as a cross-sectional ranking input).

The key insight motivating the architecture of this thesis is that these three dimensions of price are not interchangeable: they operate on different timescales, capture different economic phenomena, and, as we document empirically in Chapter 3, they exhibit predictive power in different statistical contexts. Features derived from price action and volume tend to carry cross-sectional information (which asset is performing better than others today), while features derived from the time-series behaviour of price action tend to carry time-series information (is this asset performing better than its own historical norm today). The failure to respect this distinction is, we argue, the primary reason why a single machine learning model trained on all features simultaneously underperforms the structured two-layer approach proposed in section 3.10.

This chapter reviews the theoretical foundations underlying each of these three dimensions and the portfolio construction principles used to translate signals into positions.

2.2 Cross-Sectional Momentum

The empirical foundation of this thesis rests on the momentum anomaly, one of the most robust findings in empirical asset pricing. Jegadeesh and Titman (1993) document that US equities with strong returns over the past 3 to 12 months continue to outperform low-return equities over the subsequent 3 to 12 months, generating significant risk-adjusted returns after controlling for size and value. This cross-sectional regularity (the tendency for relative winners to remain winners) has since been replicated across asset classes, geographies, and time periods (Rouwenhorst, 1998; Asness et al., 2013).

In the context of multi-asset rotation, cross-sectional momentum is implemented as a ranking procedure: at each rebalancing date, all assets in the universe are ranked by their return over a formation period, and the top subset is held for the following holding period. Asness et al. (2013) demonstrate that cross-sectional momentum generates positive returns across diverse asset classes: equities, bonds and commodities and that the momentum signal is economically independent across asset classes.

The primary quantity used to evaluate the predictive content of a cross-sectional signal is the Information Coefficient (IC), defined as the Spearman rank correlation between predicted and realized cross-sectional returns:

$$IC_t = \text{Spearman}\left(\hat{r}_{i,t},\, r_{i,t+1}\right)$$

where the correlation is computed across assets i at each time t. The mean IC and the IC Information Ratio (ICIR $= \bar{IC}/\sigma_{IC}$) are standard metrics in quantitative equity management for assessing signal quality (Grinold and Kahn, 2000). A mean IC of 0.02–0.05 is considered tradeable in large equity universes; however, as we discuss in Chapter 3, this threshold is not directly applicable to small heterogeneous universes where cross-sectional variance is compressed.

2.3 Time-Series Momentum

Barroso and Santa-Clara (2015) introduce a conceptually distinct momentum strategy: instead of ranking assets cross-sectionally against one another, time-series momentum evaluates each asset relative to its own historical return. An asset exhibits positive time-series momentum when its past 12-month return is positive, regardless of the performance of other universe constituents. Empirically, time-series momentum generates returns largely independent of cross-sectional momentum, confirming that the two signals capture fundamentally different market dynamics.

Furthermore, Barroso and Santa-Clara (2015) show that time-series momentum strategies improve significantly when position sizing is scaled inversely to realized variance, a principle

later generalised by Moreira and Muir (2017) for managing factor strategy volatility. Because momentum Sharpe ratios remain relatively stable across volatility regimes, targeting a constant risk level systematically enhances risk-adjusted performance by reducing exposure during volatile periods. A variant of this mechanism is used as a possible weighting scheme in section 3.10.2.

For multi-asset portfolios, time-series momentum is particularly advantageous because signal generation does not depend on a large cross-section. Each asset yields an independent directional signal (a positive or negative trend) irrespective of universe size. This property makes time-series signals ideally suited to the small-universe setting of this work, as formalized through the Information Coefficient analysis in Section 3.6.

2.4 Volatility as a Signal and a Risk Control

Volatility serves a dual purpose in this framework: it acts as a signal for cross-sectional asset ranking, it acts as macro-level regime filter used in machine learning models and it's used as a portfolio weighting scheme.

2.4.1 Realized Volatility as a Cross-Sectional Feature

Short-term realized volatility, calculated as the rolling standard deviation of log returns, provides both a measure of risk and predictive insight into future performance. Ang et al. (2006) first documented the idiosyncratic volatility puzzle, where assets with high idiosyncratic volatility experience anomalously low future returns, an effect widely replicated across international markets (Ang et al., 2009). In multi-asset rotation, low relative realized volatility often signals persistent trend behavior and lower drawdown risk.

We incorporate short-term realized volatility (defined as the 4-week annualized rolling standard deviation of log returns) as a cross-sectional signal, validating its predictive power through the Information Coefficient audit in Section 3.5. Beyond its ranking role, realized volatility scales position sizes through inverse-volatility weighting, adhering to the risk-parity principle that allocating equal risk is superior to allocating equal capital across heterogeneous assets (Qian, 2005; Asness et al., 2012).

2.4.2 Implied Volatility as a Regime Indicator

The CBOE Volatility Index (VIX), introduced by Whaley (1993), measures expected 30-day S&P 500 volatility implied by option pricing. Although forward-looking rather than historical, the VIX is a primary proxy for global risk sentiment, where spikes signal market stress, risk-off behavior, and heightened multi-asset correlations (Bekaert et al., 2013).

Empirical literature demonstrates that momentum strategies behave differently across market regimes. Notably, Daniel and Moskowitz (2016) show that severe momentum crashes occur predominantly in high-volatility environments following market drawdowns. Conversely, low-volatility regimes typical of sustained bull markets foster signal persistence with fewer sharp reversals.

2.5 Machine Learning in Asset Selection

The application of machine learning to cross-sectional asset selection has accelerated significantly following Gu et al. (2020), who evaluated a wide panel of firm characteristics across thousands of US stocks. Their central finding is that non-linear architectures, particularly neural networks, substantially outperform linear factor models in capturing return dynamics, achieving monthly out-of-sample R^2 values up to 0.40%.

In parallel, gradient boosted decision trees, particularly XGBoost (Chen and Guestrin, 2016), have become a standard benchmark in quantitative finance. Their strength lies in capturing non-linear feature interactions, providing robust regularization against noisy variables, and operating efficiently on tabular datasets. For example, Fischer and Krauss (2018) applied gradient boosted trees to S&P 500 constituents rotation, documenting out-of-sample Sharpe ratios around 0.5 when relying on return-based signals.

However, a key condition for the success of these architectures is the size and homogeneity of the underlying cross-section. Large equity universes provide thousands of comparable observations per date, whereas a small multi-asset universe comprising equity sectors, fixed income, and commodities yields only 20 to 30 assets per rebalance. In such constrained settings, highly flexible models struggle to identify stable cross-sectional interactions without overfitting to sample noise, a structural limitation evaluated empirically in Section 3.11-12.

Finally, applying machine learning models to sequential financial data requires rigorous evaluation protocols. Standard k-fold cross-validation breaks temporal ordering and introduces look-ahead bias, generating artificially inflated backtest performance (López de Prado, 2018). Evaluating predictive models via expanding-window walk-forward validation ensures that model training and signal generation rely strictly on information available at each historical decision point.

2.6 Volume-Based Price Levels: The Point of Control

While price returns capture directional trajectory, volume distributions reveal structural market acceptance. The Point of Control (POC), introduced in Market Profile theory (Steidlmayer and Koy, 1984), represents the price level at which the largest trading volume accumulated within

a specified lookback window. Conceptually, it serves as a volumetric centre of gravity where market participants collectively transacted most frequently.

Unlike conventional price momentum, tracking shifts in volume concentration distinguishes genuine positional accumulation from low-volume price momentum. An upward migration in volume-weighted fair value provides an early cross-sectional signal of institutional interest, rendering volume profile metrics complementary to traditional price-based indicators.

2.7 The Kalman Filter as a Fair-Value Estimator

The Kalman filter (Kalman, 1960) is a recursive Bayesian estimator that computes the minimum mean squared error estimate of an unobserved state from noisy measurement data. When applied to financial price series, it models the underlying trend as a hidden state continuously updated by incoming price observations.

Unlike static moving averages, the Kalman filter dynamically updates its gain parameter based on local signal-to-noise ratio and process variance. During high-volatility regimes, the filter automatically adapts its weighting to accommodate rapid price discovery, whereas during quiet regimes, it smooths out transient noise. This adaptive property makes it particularly suited as a dynamic fair-value benchmark: deviations between the observed price and the filtered state capture short-term mispricings without suffering from the structural lag inherent in standard moving averages.

2.8 Portfolio Construction in Multi-Asset Settings

The construction of a portfolio from a ranked set of assets involves two decisions: which assets to hold, and in what proportions. The second has been extensively studied under mean-variance optimisation (Markowitz, 1952), which selects weights to maximise expected return for a given level of risk. A well-known limitation on small portfolios is the sensitivity of optimal weights to estimation error in the covariance matrix (Michaud, 1989).

López de Prado (2016) proposes Hierarchical Risk Parity (HRP), which constructs weights by applying Ward's linkage clustering to the correlation matrix and allocating risk inversely proportional to within-cluster variance, avoiding matrix inversion. Marchenko-Pastur denoising (Bouchaud and Potters, 2009), removes noise eigenvalues from the sample covariance matrix before optimisation, improving conditioning on short estimation windows.

In this thesis we evaluate five construction schemes: Equal Weight, Inverse Volatility, Score Weighted, Equal Risk Contribution, and HRP. The finding that all methods produce similar Sharpe ratios on a three-asset portfolio confirms the theoretical result of Grinold and Kahn (2000): for sufficiently concentrated portfolios, signal quality dominates weighting precision.

3. Methodology

3.1 Data & Universe

The investable universe consists of 21 exchange-traded funds (ETFs) listed on US exchanges, covering five distinct asset classes: US equity sectors, international equity indices, fixed income, real assets, and currencies. Table 3.1 provides a full description.

3.1.1 The 21-ETF Multi-Asset Universe

Table 3.1: The 21-ETF investable universe.

Ticker	Name	Asset Class	Category
XLK	Technology Select Sector SPDR	US Equity	Sector
XLY	Consumer Discr. Select Sector SPDR	US Equity	Sector
XLF	Financial Select Sector SPDR	US Equity	Sector
XLV	Health Care Select Sector SPDR	US Equity	Sector
XLI	Industrials Select Sector SPDR	US Equity	Sector
XLP	Consumer Staples Select Sector SPDR	US Equity	Sector
XLU	Utilities Select Sector SPDR	US Equity	Sector
XLE	Energy Select Sector SPDR	US Equity	Sector
XLB	Materials Select Sector SPDR	US Equity	Sector
IWM	iShares Russell 2000 ETF	US Equity	Index
DIA	SPDR Dow Jones Industrial ETF	US Equity	Index
EFA	iShares MSCI EAFE ETF	International Equity	Index
EEM	iShares MSCI Emerging Markets ETF	International Equity	Index
EWJ	iShares MSCI Japan ETF	International Equity	Index
TLT	iShares 20+ Year Treasury Bond ETF	Fixed Income	Bond
IEF	iShares 7-10 Year Treasury Bond ETF	Fixed Income	Bond
LQD	iShares iBoxx $ Investment Grade Corporate Bond ETF	Fixed Income	Bond
HYG	iShares iBoxx $ High Yield Corporate Bond ETF	Fixed Income	Bond
TIP	iShares TIPS Bond ETF	Fixed Income	Bond
GLD	SPDR Gold Shares	Real Assets	Commodity
VNQ	Vanguard Real Estate ETF	Real Assets	REIT

The universe is designed to span the major risk premia available to a multi-asset investor (equity risk, duration risk, credit risk, inflation protection, and real asset exposure) while remaining constrained to liquid, exchange-traded instruments with full price and volume history available from April 2007. Several instruments considered in earlier iterations of this work were excluded: single-commodity ETFs (USO, SLV, DBA) due to their predominantly mean-reverting behaviour and low trend persistence on weekly frequency, which was confirmed in the feature IC audit. FXI

(iShares China Large-Cap) was also excluded due to structural breaks in liquidity and market access restrictions during the sample period.

SPY (SPDR S&P 500 ETF) is used exclusively as a benchmark for performance attribution and is not part of the investable universe. The CBOE Volatility Index (VIX) is used as a macro regime indicator for portfolio exposure scaling and is sourced separately via daily data resampled to weekly frequency.

3.1.2 Data Sources and Sample Period

Price data for all 21 universe constituents and SPY are sourced from TradingView historical exports in OHLCV format (Open, High, Low, Close, Volume) at daily frequency. VIX daily data is sourced from Yahoo Finance. All series share a common sample period of **April 2007 to May 2026**, providing a full history of 998 weekly observations.

The sample period was selected to satisfy three criteria. First, all 21 ETFs must have live price and volume history throughout the entire window — this excludes newer instruments and constrains the start date to April 2007. Second, the period must include at least one full market cycle to allow the feature IC audit to distinguish genuine predictive relationships from those specific to a single regime. The sample encompasses the Global Financial Crisis (2008–2009), the European Sovereign Debt Crisis (2011–2012), the COVID-19 crash and recovery (2020), the 2022 rate-hiking cycle, and the AI-driven bull market of 2023–2026. Third, the out-of-sample period (2009–2026, 893 weeks) must be long enough to produce statistically meaningful performance estimates, requiring at least 300 weekly observations for the t-statistics reported in sections 3.5-6.

3.1.3 Data Processing

3.1.4 Weekly Resampling

All daily OHLCV data is resampled to weekly frequency using Friday-end (W-FRI) bars:

- **Open**: first daily open of the week
- **High**: maximum daily high of the week
- **Low**: minimum daily low of the week
- **Close**: last daily close of the week (Friday or last trading day)
- **Volume**: sum of daily volumes over the week

Weekly frequency is chosen for three reasons. First, it reduces microstructure noise that affects daily signals without losing the medium-term trend information captured by the features. Second, it produces a manageable turnover rate for a rotational strategy targeting liquid ETFs. Third, the feature lookback windows used in this thesis (4w, 13w, 26w, 52w) correspond to natural investment horizons of one month, one quarter, six months, and one year respectively.

3.1.5 Alignment and Missing Data

All series are aligned to a common weekly index — the intersection of all weekly dates across the 21 universe constituents. Missing values arising from ETF-specific holidays or data gaps are forward-filled using the most recent available close, a standard procedure for liquid ETF data where gaps are almost always due to calendar differences rather than genuine trading halts. The final aligned dataset contains 998 weekly observations with no remaining missing values across all 21 constituents.

3.1.6 Return Computation

Weekly log returns are computed as:

$$r_{i,t} = \ln\left(\frac{p_{i,t}}{p_{i,t-1}}\right)$$

where $p_{i,t}$ is the Friday close of asset i in week t. Log returns are used throughout the feature engineering and performance measurement stages for their additivity over time and their approximate symmetry in tails. Portfolio-level performance statistics reported in Chapter 6 are converted to simple (arithmetic) returns for compounding.

3.2 Descriptive Statistics

Table 3.2 reports annualised return, volatility, and Sharpe ratio for all 21 universe constituents over the full sample (April 2007 – May 2026), together with the benchmark SPY.

Note: Sharpe ratio computed as annualised return divided by annualised volatility. No risk-free rate deducted for comparability across instruments.

Several observations are noteworthy. First, the return dispersion across asset classes is substantial reflecting the heterogeneity of the universe that makes cross-sectional ML models difficult to train. Second the wide volatility ranges across assets makes equal-weight portfolio construction sub-optimal from a risk contribution perspective. Third, SPY's Sharpe over this period reflects the exceptional nature of the US equity bull market driven by quantitative easing and the

Table 3.2: Descriptive statistics, April 2007 – May 2026.

Asset	Ann. Return	Ann. Vol	Sharpe	Max Drawdown	Category
XLK	19.8%	19.2%	1.03	-54.1%	Sector
XLY	14.6%	18.8%	0.78	-55.2%	Sector
XLV	12.3%	14.1%	0.87	-32.4%	Sector
XLF	8.4%	23.9%	0.35	-79.2%	Sector
XLI	11.8%	17.9%	0.66	-52.1%	Sector
XLP	9.7%	11.8%	0.82	-28.3%	Sector
XLU	8.9%	14.6%	0.61	-40.2%	Sector
XLE	5.3%	23.4%	0.23	-62.0%	Sector
XLB	10.2%	18.8%	0.54	-55.8%	Sector
IWM	10.1%	19.8%	0.51	-57.8%	Equity Index
DIA	11.3%	16.4%	0.69	-51.9%	Equity Index
EFA	5.8%	16.9%	0.34	-57.5%	Int'l Equity
EEM	5.1%	20.7%	0.25	-65.2%	Int'l Equity
EWJ	4.9%	16.3%	0.30	-51.3%	Int'l Equity
TLT	2.8%	14.1%	0.20	-46.2%	Bond
IEF	3.2%	7.8%	0.41	-21.8%	Bond
LQD	4.1%	8.2%	0.50	-24.3%	Bond
HYG	4.8%	9.4%	0.51	-33.2%	Bond
TIP	3.4%	6.1%	0.56	-19.4%	Bond
GLD	8.1%	15.9%	0.51	-42.9%	Real Asset
VNQ	7.2%	20.1%	0.36	-72.3%	Real Asset
SPY	**13.5%**	**16.3%**	**0.82**	**-55.2%**	Benchmark

technology sector rally — the benchmark against which our strategy is measured is therefore unusually strong relative to any multi-asset alternative.

3.3 Walk-Forward Validation and Dataset Split

Standard cross-validation techniques such as k-fold splitting are fundamentally flawed when applied to financial time series, as shuffling observations violates temporal ordering and introduces look-ahead bias (López de Prado, 2018). To enforce strict causality, at any rebalancing date t all signal computation and portfolio construction rely strictly on information available up to $t-1$.

The complete dataset spans April 2007 – May 2026 across 21 liquid ETFs and is partitioned into three operational periods, detailed in Table 3.3.

The warm-up length differs slightly across model variants: the dual-layer rule-based system requires approximately 150 weeks (`SEED_WEEKS`) to initialise its 52-week rolling z-scores, while the LambdaRank ML model requires an additional burn-in period of approximately 200 weeks to accumulate sufficient training observations before the first ranking model is fitted. Consequently, strategy returns are available from approximately mid-2010 for the dual-layer system and from early 2011 for the ML model.

Table 3.3: **Dataset partitioning and walk-forward validation structure.**

Period	Dates	Weeks	Operational Purpose
Warm-up (Seed)	Apr 2007 – 2010/2011	150–200	Rolling z-score initialisation; no allocations generated
In-Sample (IS)	2010/2011 – Dec 2021	$\sim$570–620	Feature IC audit & architecture calibration
Pure OOS Evaluation	Jan 2022 – May 2026	232	Unseen regime validation; frozen parameters

This partitioning structure separates two core empirical objectives:

1. **Feature Selection (In-Sample):** The IC audit and redundancy screening are conducted strictly over the 2007–2021 period (`AUDIT_END_DATE`). Concluding design in 2021 prevents look-ahead leakage from the 2022+ rate-hiking cycle, which represents a structurally distinct macroeconomic regime relative to the post-GFC low-rate environment dominating the IS window.
2. **Strategy Execution (Walk-Forward):** Both strategies generate signals sequentially in a walk-forward fashion, iterating over each rebalancing date t and computing all inputs exclusively from data available up to $t-1$. The dual-layer rule-based system applies fixed, economically-motivated weights at each rebalancing date with no periodic model refitting. While execution spans 2010/2011–2026, the **January 2022 – May 2026 window constitutes the pure out-of-sample test**, as it lies entirely beyond the feature audit horizon.

We acknowledge that the execution period 2010/2011–2021 overlaps with the feature audit window, which warrants methodological clarification. Feature selection was not based on optimizing portfolio-level performance metrics over this period, but was limited to confirming the statistical validity (via information coefficient t-statistics) of factors already established in the cross-sectional asset pricing literature (Jegadeesh and Titman, 1993). The dual-layer scoring architecture employs fixed, economically-motivated weights rather than estimated coefficients, and no in-sample optimization of combination rules or portfolio construction parameters was performed. Signal normalization relies exclusively on expanding z-scores computed at each rebalancing date t using only data up to $t-1$, ensuring that no future information contaminates the ranking process at any point.

Nevertheless, we conservatively designate January 2022 onward as the **pure out-of-sample** evaluation window. This window spans four years encompassing the 2022 rate-hiking cycle, the subsequent disinflation period, and the 2025 tariff-driven volatility episode (regime conditions entirely absent from the IS calibration environment). Both the dual-layer and ML strategies achieve **Sharpe ratios 50-60%** higher than the SPY benchmark over this window, providing strong empirical evidence against in-sample overfitting.

3.4 Feature Engineering & Validation

3.4.1 Overview

Feature engineering in quantitative finance is the process of transforming raw price, volume, and volatility data into numerical signals that carry predictive information about future returns. The quality of any downstream model, whether a rank-based filter or a machine learning model, is bounded by the quality of its input. A feature without predictive power cannot contribute to portfolio alpha regardless of model complexity; conversely, a feature with strong predictive power can generate alpha even when combined in a simple linear ranking rule.

This chapter documents the construction of six features (two from each of the three price dimensions introduced in Chapter 2) and their systematic validation using the Information Coefficient framework. The validation proceeds in two distinct contexts: cross-sectional (comparing assets to each other) and time-series (comparing each asset to its own history). As we demonstrate empirically, these two contexts produce systematically different results for the same feature, a finding that motivates the two-layer architecture of Section 3.10.

All features are computed on the weekly OHLCV data. All IC statistics are computed exclusively on the in-sample period to ensure that no feature selection decision uses out-of-sample data.

3.4.2 Feature Construction

3.4.3 Price Action Features

Kalman Filter Deviation (`kalman_dev`) Building on the adaptive framework reviewed in Chapter 2, we model the intrinsic fair value $\mu_{i,t}$ of asset i at time t as a random walk observed through noisy price measurements $p_{i,t}$:

$$\mu_{i,t} = \mu_{i,t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q) \qquad p_{i,t} = \mu_{i,t} + v_t, \quad v_t \sim \mathcal{N}(0, R) \tag{3.1}$$

where Q and R represent the process and measurement variances, respectively. The recursive update yields a smooth fair-value estimate $\hat{\mu}_{i,t}$, from which the normalized deviation feature is defined as:

$$\text{kalman_dev}_{i,t} = -\frac{p_{i,t} - \hat{\mu}_{i,t}}{\hat{\mu}_{i,t}} \tag{3.2}$$

The negative sign ensures that a positive value represents an asset trading *below* its Kalman fair value, signaling a potential mean-reversion buying opportunity within an established trend.

Implementation parameters are set to $Q = 10^{-4}$ and $R = 10^{-2}$. These values yield a steady-state

Kalman gain $K \approx 0.09$, updating the fair-value estimate by 9% of the innovation at each step and producing a smooth baseline with an effective half-life of roughly 7.5 weeks.

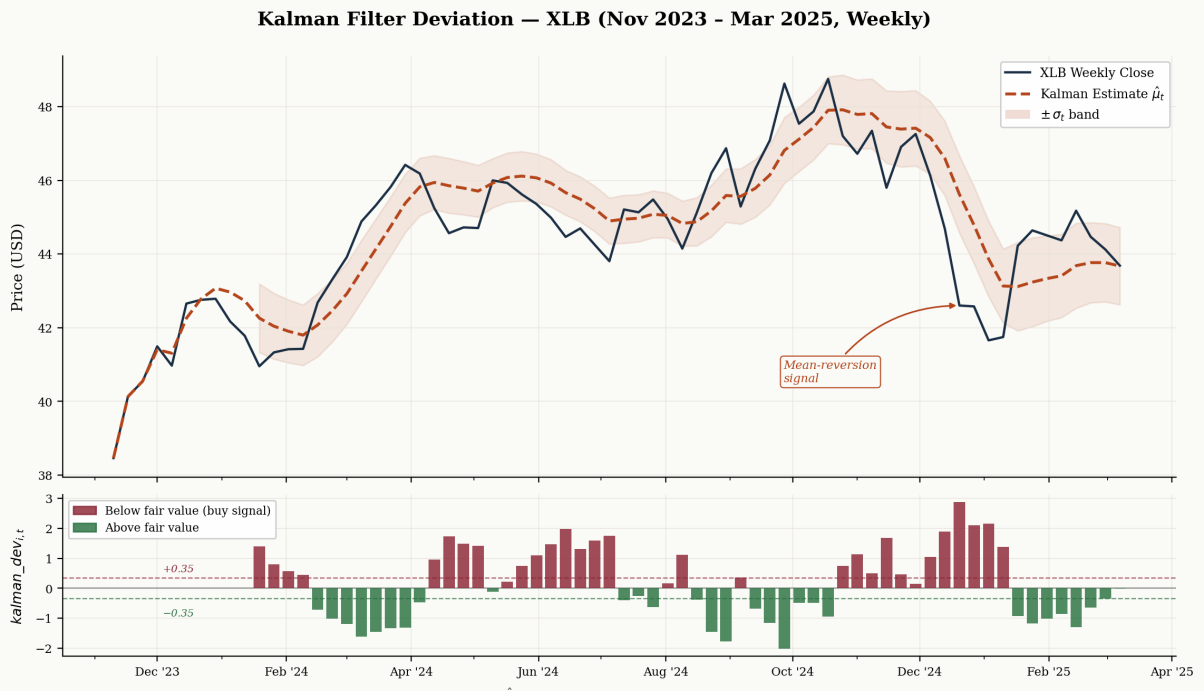


Figure 3.1: **Kalman filter applied to a weekly price series.** The upper panel displays the raw close price alongside the adaptive fair value estimate $\hat{\mu}_t$. The lower panel reports kalman_dev$_{i,t}$ as defined in Equation 3.2, where positive values highlight assets trading below fair value.

52-Week Momentum (`mom_52w`) FollowingMoskowitz et al. (2012), the 52-week momentum feature is defined as:

$$\text{mom_52w}_{i,t} = \frac{p_{i,t-1}}{p_{i,t-53}} - 1 \tag{3.3}$$

The one-week skip (using p_{t-1} rather than p_t) avoids the short-term reversal effect documented by Jegadeesh and Titman (1993), which would otherwise contaminate the medium-term momentum signal with negative autocorrelation at lag 1. This is the canonical Moskowitz formation period and the only momentum lookback that survives the cross-sectional IC audit at the $|t| \geq 2$ threshold.

Trend Exhaustion (`sma_deriv_neg`)

The trend exhaustion feature is derived from the second-order behaviour of the price trend. We first compute the derivative of the 12-week simple moving average:

$$\text{sma_deriv}_{i,t} = \text{SMA}(p_i, 12)_t - \text{SMA}(p_i, 12)_{t-3} \tag{3.4}$$

This measures the rate of change of the medium-term trend over the past three weeks. The

feature is then negated: $trend_exhaustion_{i,t} = -sma_deriv_{i,t}$.

The rationale for negation is established empirically in Section 3.4: in the time-series context, a rapidly accelerating trend (high sma_deriv) tends to be followed by underperformance, consistent with an overextension or trend exhaustion effect. The negated feature captures the regime where the trend is decelerating relative to the asset's own history, which is associated with a higher probability of continued positive returns. This finding is not present in the cross-sectional context (where the feature has near-zero IC), but is strong and stable in the time-series context.

3.4.4 Volume Features

POC Cross (`poc_cross`) Building on the volume-profile principles outlined in Chapter 2, the Point of Control cross measures the relative shift between short-term ($50w$) and long-term ($78w$) volume-weighted price consensus:

$$\text{poc_cross}_{i,t} = \frac{\text{POC}_{i,t}^{50w} - \text{POC}_{i,t}^{78w}}{p_{i,t}} \tag{3.5}$$

where $\text{POC}_{i,t}^{W}$ represents the rolling Fixed-Range Volume Profile Point of Control over a rolling window of W weeks. A positive $\text{poc_cross}_{i,t}$ signifies an upward shift in recent volume accumulation relative to the structural baseline, indicating a bullish migration in market consensus.

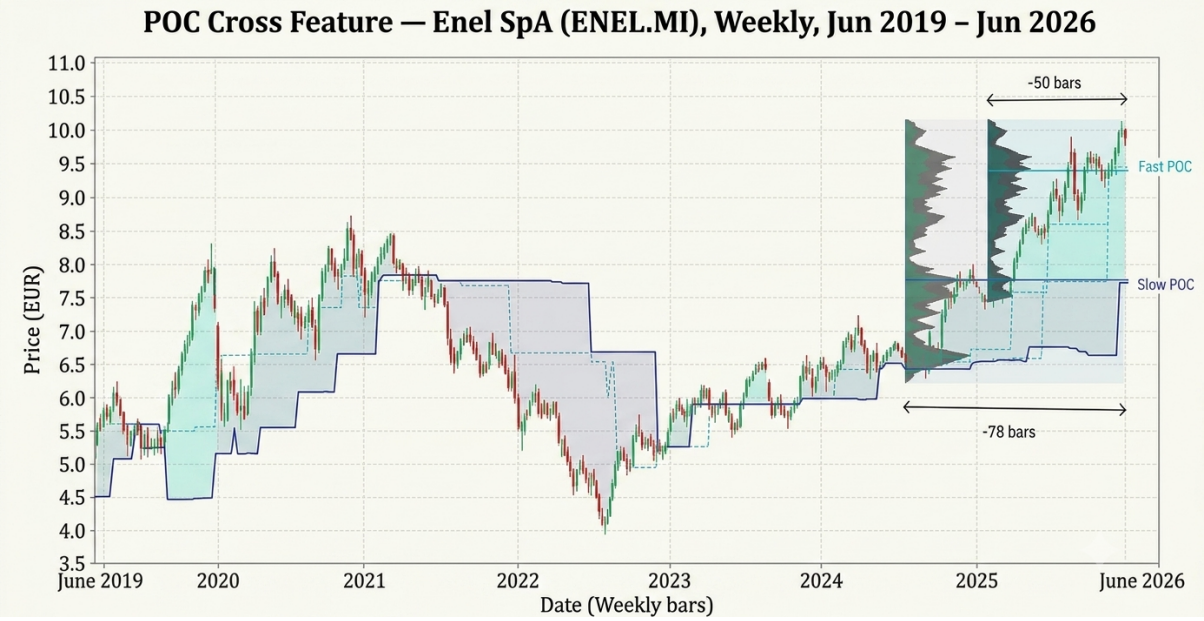


Figure 3.2: **Illustration of `poc_cross` on a weekly price chart.** The short-term POC ($50w$, dashed) tracks recent volume concentration, while the long-term POC ($78w$, solid) acts as a structural reference.

Computational implementation utilizes a histogram approach on OHLCV data. For each weekly bar, the price range $[\text{Low}_t, \text{High}_t]$ is discretized into price bins of width $\Delta p = 0.1\% \cdot p_t$. Volume

is allocated proportionally across overlapping bins weighted by bar volume, with the final POC identified as the bin accumulating highest total volume over the rolling window.

3.4.5 Volatility Features

Short-Term realized Volatility (`vol_4w`)

Short-term realized volatility is computed as the rolling standard deviation of weekly log returns over a 4-week window, annualised:

$$\text{vol_4w}_{i,t} = \sqrt{52} \cdot \sigma\left(r_{i,t-3}, r_{i,t-2}, r_{i,t-1}, r_{i,t}\right) \tag{3.6}$$

where $r_{i,t} = \ln(p_{i,t}/p_{i,t-1})$ is the weekly log return. The 4-week window captures very recent volatility dynamics — reflecting the current risk environment for the asset — rather than structural long-run volatility which changes slowly. A minimum of 2 observations is required; if fewer are available, the feature is marked as missing and the asset is excluded from the ranking that week.

2-Week Short-Term Reversal (`rev_2w`)

The two-week short-term reversal feature is defined as:

$$\text{rev_2w}_{i,t} = -\left(\frac{p_{i,t}}{p_{i,t-2}} - 1\right) \tag{3.7}$$

This metric captures the short-term mean-reversion dynamics documented by Jegadeesh and Titman (1993), where assets experiencing sharp recent losses exhibit temporary performance rebounds. The negative sign ensures that a higher value corresponds to a recent under-performer, signaling a potential mean-reversion buying opportunity. This feature is validated in Section 3.8, where the alpha-orthogonality test evaluates its incremental predictive contribution relative to the Kalman filter deviation.

3.4.6 Example: Why the Same Feature Behaves Differently in the Two Contexts

To build intuition before presenting the formal results, we illustrate the difference between cross-sectional and time-series IC evaluation using a stylised example with three assets: XLK (technology equity), TLT (long-term bonds), and GLD (gold), observed over four consecutive weeks. All values are hypothetical but order-of-magnitude realistic.

Setup: Observed Prices

Table 3.4: Observed prices over the 4-week window (and 52-week anchor).

	XLK	TLT	GLD
Week -48 (52w prior)	75.7	82.6	143.0
Week 1	100	80	150
Week 2	108	79	152
Week 3	106	77	155
Week 4	112	76	153
`mom_52w` at Week 4	$+48.0\%$	-8.0%	$+7.0\%$

We use `mom_52w` as the illustrative feature, with hypothetical trailing 52-week prices such that the momentum values at Week 4 are:

$$\text{XLK} = +48\%, \quad \text{TLT} = -8\%, \quad \text{GLD} = +7\% \tag{3.8}$$

Cross-Sectional IC Evaluation at Week 4

In the cross-sectional test, we ask: *among the three assets this week, does a higher `mom_52w` predict which asset will perform best next week?*

We rank the assets by `mom_52w` in ascending order (rank 1 = lowest momentum):

Table 3.5: Cross-sectional ranking example at Week 4.

Asset	mom_52w	CS Rank	Fwd Return (Week 5)	Return Rank
TLT	-8.0%	1	-1.5%	1
GLD	$+7.0\%$	2	$+1.0\%$	2
XLK	$+48.0\%$	3	$+3.2\%$	3

The Spearman correlation between the `mom_52w` rank $[1,2,3]$ and the forward return rank $[1,2,3]$ is $+1.0$, a perfect positive correlation. The asset with the highest cross-sectional momentum (XLK) delivers the highest return, while the asset with the lowest (TLT) delivers the worst. This is the cross-sectional effect: *relative winners continue to be relative winners*.

Time-Series IC Evaluation for XLK

In the time-series test, we ask a completely different question: *when XLK's `mom_52w` is unusually high relative to XLK's own historical distribution, does XLK tend to perform well the following week?*

XLK's rolling 52-week momentum mean and standard deviation evolve gradually as new data enters the estimation window. The resulting z-scored values are reported below:

Table 3.6: Time-series z-score example for XLK.

Week	XLK mom_52w	Rolling Mean	Rolling Std	z-score	Fwd Return
1	$+12\%$	$+14.5\%$	16.0%	-0.16	$+8.0\%$
2	$+28\%$	$+15.0\%$	16.2%	$+0.80$	-1.9%
3	$+38\%$	$+15.5\%$	16.5%	$+1.36$	$+5.7\%$
4	$+48\%$	$+16.0\%$	16.8%	$+1.90$	$+3.2\%$

Computing Spearman(z-score, fwd return) over these four observations, the z-score ranks $[1,2,3,4]$ map to forward return ranks $[1,3,4,2]$, yielding a correlation of approximately -0.20. Across a longer history and all 21 assets, this negative relationship is stable and significant ($t = -3.49$, Table 3.9).

The economic interpretation is the following. At Week 2, XLK's momentum had accelerated to 0.8σ above its own historical norm, and the following week the return was -1.9%. At Week 4, momentum reached 1.9σ above norm, an overextended reading, yet the return, while positive, was the second smallest in the sample. When a trend has accelerated far beyond an asset's own historical norm, the probability of continuation at that pace diminishes. This is the *trend exhaustion* phenomenon that the negated `sma_deriv` feature is designed to capture.

Core Insights from the Numerical Example

The same feature, `mom_52w`, tells two different stories depending on the context in which it is evaluated:

- **Cross-sectionally:** XLK has higher momentum than TLT and GLD this week, therefore XLK will outperform them." This is correct and consistent with the Moskowitz (2012) cross-sectional momentum result.
- **Time-series:** XLK's momentum is 1.90σ above XLK's own historical norm, therefore XLK will outperform its own average next week." This is *incorrect*: the overextension signal predicts underperformance, not continuation.

A machine learning model trained on a cross-sectional panel receives both signals simultaneously for every asset-week observation. Because the cross-sectional signal is positive and the time-series signal is negative, the model receives conflicting gradients and learns a compromise that is weaker than either signal used in isolation. The two-layer architecture proposed in section 3.10 resolves this conflict by assigning each feature to the layer where its predictive direction is supported by the IC evidence.

3.4.7 Extended Candidate Pool and Signal Normalization

Rather than designing the two-layer architecture around the six features of Section 3.4 in isolation, the selection protocol is deliberately data-driven: those six features are evaluated *jointly*, on equal footing, against an unconstrained pool of 15 candidate signals spanning price, volume, and volatility dynamics.

A crucial methodological step, before any evaluation can take place, involves the homogenization of the input signals. To ensure statistical comparability and mathematical validity within the sequential ranking funnel, all 15 candidate features are subjected to a centralized rolling time-series Z-score normalization pipeline. This transformation anchors each raw signal to its own statistical baseline, centering its distribution locally at zero with unit variance. For each feature k, asset i, and week t, the rolling Z-score $z_{k,i,t}$ is computed as follows:

$$z_{k,i,t} = \frac{x_{k,i,t} - \mu_{i,t}^{(52)}}{\sigma_{i,t}^{(52)}} \tag{3.9}$$

where $x_{k,i,t}$ represents the raw feature value, and $\mu_{i,t}^{(52)}$ and $\sigma_{i,t}^{(52)}$ are, respectively, the rolling mean and standard deviation calculated over a backward-looking window of $N = 52$ weeks for that specific asset. This one year lookback period ensures sufficient statistical robustness for the local parameter estimation while capturing annual market cyclicity. To prevent numerical instability due to outliers, the resulting z-scores are clipped to the range $[-3, 3]$.

This normalization process is an important methodological point that shapes how the remainder of this chapter should be read. The six features introduced above are not a starting assumption that the audit merely confirms; they are the six survivors of an audit conducted across all 15 normalized candidates, which is why they alone were given a full economic and formulaic treatment in Section 3.4. The audit itself, however, is run once, on the full standardized pool, and the architectural decision in Section 3.9 follows directly from its results.

Table 3.7 lists all 15 candidates together with their dimension of origin, economic rationale, and defining formula.

Table 3.7: **Extended candidate pool: 15 features in their final z-scored pipeline representation.** All features are standardized using a 52-week rolling window, $Z_{52w}(\cdot)$, and clipped to $[-3,3]$ unless specified otherwise.

Feature	Signal Interpretation & Reference	Normalized Formula (z-score)
Already defined features (Section 3.9)		
kalman_dev, poc_cross, sma_deriv_neg, vol_4w, mom_52w, and rev_2w are recalled here by reference, subjected to the same 52-week rolling z-score normalization.		
New candidates (9)		
vol_26w	26-week realized volatility (Moreira and Muir, 2017). Captures slow-moving volatility regimes, smoothing transient weekly spikes in vol_4w.	$Z_{52w}\left(\text{vol}_{26w,t}\right)$
mom_26w	6-month momentum (Jegadeesh and Titman, 1993). Tests whether intermediate lookbacks offer incremental ranking power over the 12-month window.	$Z_{52w}\left(p_{t-1}/p_{t-27}-1\right)$
calmar_neg_26w	Negated rolling Calmar ratio over 26 weeks. Measures risk-adjusted trend quality; strong trends with minimal drawdowns are negated to test for exhaustion.	$Z_{52w}\left(-\frac{p_t/p_{t-26}-1}{\lvert\min_{s\le t}(p_s/\max_{u\le s}p_u-1)\rvert+\varepsilon}\right)$
pearson_r_neg	Negated 26-week Pearson correlation between price and a linear time index. Quantifies trend linearity, where values near $+1$ represent an ideal trend.	$Z_{52w}(-\text{corr}(\{p_{t-25},\ldots,p_t\},\ \{1,\ldots,26\}))$
f_poc_xs	Inverted, time-series normalized distance from 16-week POC. Serves as a short-window volume-based mean-reversion signal.	$-Z_{52w}\left(\frac{p_t-\text{POC}_{16w}}{\text{POC}_{16w}}\right)$
sma50_zscore	Negated distance of price from the 50-week SMA, normalized by rolling volatility. Analogous to kalman_dev with a deterministic reference.	$-Z_{52w}\left(\frac{p_t-\text{SMA}_{50w}}{\sigma_{\text{SMA}_{50w}}+\varepsilon}\right)$
volume_zscore	Abnormal trading volume relative to historical activity. Microstructure feature distinct from price-derived candidates.	$Z_{52w}\left(V_t\right)$
vix_zscore	VIX z-scored against a 252-week window. Macro risk-regime indicator evaluated for time-series regime timing rather than cross-sectional ranking.	$Z_{252w}\left(\text{VIX}_t\right)$
price_range_z	Intraweek high-low range relative to close price. Captures intraperiod dispersion not present in close-to-close returns.	$Z_{52w}\left(\frac{H_t-L_t}{p_t}\right)$

The audit proceeds in three stages, mirroring the structure of the remainder of this chapter: (i) a dual-context Information Coefficient audit, testing each candidate cross-sectionally and in time-series (Sections 3.5–3.6.3); (ii) a pairwise redundancy map across all $\binom{15}{2}=105$ candidate pairs (Section 3.7); and (iii) an alpha-orthogonality test that converts raw correlation into a statement about incremental predictive content (Section 3.8). The final architecture in Section 3.9

is the direct output of this three-stage funnel, not a predefined structure the audit is built to confirm.

3.5 Cross-Sectional IC Audit

3.5.1 Methodology

The cross-sectional IC audit measures whether each feature can rank assets in the correct order relative to each other in the same week. Formally, for each week t in the in-sample period, we compute the Spearman rank correlation between the feature values across all 21 assets and their forward one-week log return:

$$IC_t^{CS} = \text{Spearman}\left(\{f_{i,t}\}_{i=1}^{N},\ \{r_{i,t+1}\}_{i=1}^{N}\right) \tag{3.10}$$

The time series $\{IC_t^{CS}\}$ is then summarised by its mean, standard deviation, and Information Ratio:

$$\bar{IC}^{CS} = \frac{1}{T}\sum_{t=1}^{T} IC_t^{CS}, \qquad ICIR^{CS} = \frac{\bar{IC}^{CS}}{\sigma_{IC^{CS}}} \tag{3.11}$$

Statistical significance is assessed via the t-statistic:

$$t^{CS} = \frac{\bar{IC}^{CS}}{\sigma_{IC^{CS}}/\sqrt{T}} \tag{3.12}$$

under the null hypothesis $H_0 : \bar{IC}^{CS} = 0$. A threshold of $|t| \geq 2.0$ is used throughout this chapter as the bar for statistical significance at approximately the 95% confidence level.

An important design principle: the IC audit is conducted on the **full universe of 21 assets with no pre-filtering**. Pre-filtering assets before the audit (for example, restricting to the top-10 by momentum before computing cross-sectional IC) compresses the cross-sectional variance of the feature, artificially deflating both the IC and its t-statistic. All 15 candidates are evaluated on the identical cross-section at each date.

3.5.2 Results

Table 3.8 reports the cross-sectional IC statistics for all 15 candidates over the in-sample period (April 2007 – December 2021).

Only two candidates clear the $|t| \geq 2.0$ significance bar cleanly: `kalman_dev` ($t = +2.66$) and

Table 3.8: Cross-sectional IC audit, full 15-candidate pool (April 2007 – December 2021).

Feature	Mean IC	Std IC	ICIR	t-stat	IC > 0 (%)	Verdict
kalman_dev	+0.0374	0.387	+0.097	**+2.66**	53.8%	Strong
poc_cross	+0.0261	0.260	+0.101	**+2.64**	54.2%	Strong
mom_52w	+0.0327	0.450	+0.073	+1.94	53.1%	Good
vol_4w	+0.0255	0.403	+0.063	+1.75	52.1%	Good
f_poc_xs	+0.0229	0.369	+0.062	+1.67	50.8%	Good
vol_26w	+0.0270	0.467	+0.058	+1.59	52.1%	Good
mom_26w	+0.0197	0.449	+0.044	+1.19	52.6%	Weak
rev_2w	+0.0127	0.413	+0.031	+0.85	50.3%	Noise
volume_zscore	+0.0015	0.280	+0.005	+0.14	51.3%	Noise
price_range_z	-0.0015	0.342	-0.004	-0.12	50.0%	Noise
sma50_zscore	-0.0029	0.406	-0.007	-0.20	50.1%	Noise
sma_deriv_neg	-0.0091	0.423	-0.021	-0.59	48.1%	Noise
calmar_neg_26w	-0.0096	0.420	-0.023	-0.62	49.5%	Noise
pearson_r_neg_26w	-0.0108	0.413	-0.026	-0.72	49.3%	Noise

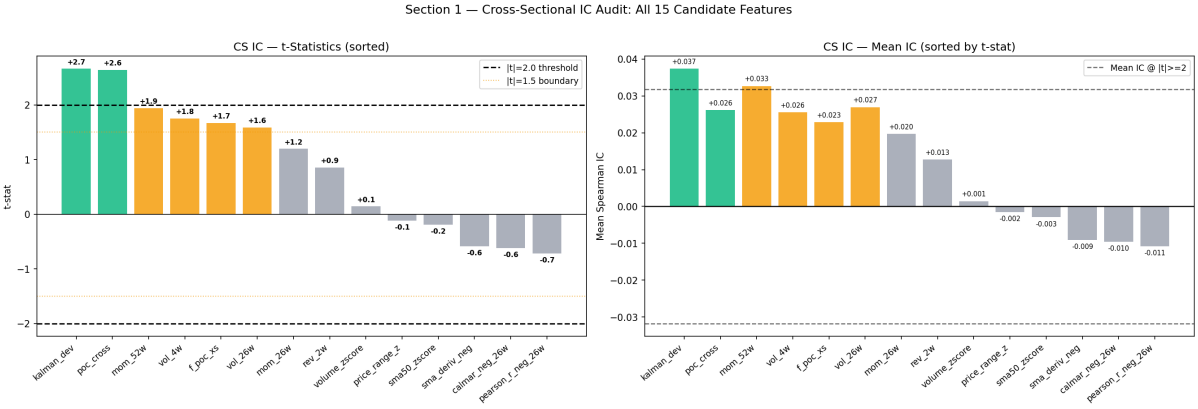


Figure 3.3: **Cross-sectional IC audit, full 15-candidate pool.**

`poc_cross` ($t = +2.64$). `mom_52w` falls just short ($t = +1.94$) but is retained as the canonical Moskowitz et al. (2012) cross-sectional momentum signal on economic grounds, discussed further in Section 3.9. A second tier of candidates – `vol_4w`, `f_poc_xs`, and `vol_26w` – shows weaker but directionally consistent cross-sectional signal ($1.5 \leq t < 2.0$); as shown in Section 3.7, two of these three are ultimately excluded on redundancy grounds rather than on IC alone. The remaining candidates, including `sma_deriv_neg` ($t = -0.59$), show no cross-sectional predictive power – a result that is not a weakness of these features but rather the first half of the context-dependency finding developed in Section 3.6.3.

The ICIR values, while modest in absolute terms (0.06–0.10 for the two strongest features), are consistent with published results on multi-asset universes: Asness, Moskowitz & Pedersen (2013) report monthly ICs of approximately 0.04–0.07 for cross-asset momentum, translating to weekly ICs of 0.01–0.03 under square-root-of-time scaling.

3.6 Time-Series IC Audit

3.6.1 Methodology

The time-series IC audit measures a fundamentally different quantity: for each individual asset, does a feature predict that asset's own future return when evaluated relative to the asset's own historical distribution?

The time-series IC is defined analogously to equation (3.9), but computed within each asset over time rather than across assets at each point in time. clipped at ± 3 to limit outlier influence. For each asset i, the time-series IC is defined as:

$$IC_i^{TS} = \text{Spearman}\left(\{z_{i,t}\}_{t=1}^{T},\ \{r_{i,t+1}\}_{t=1}^{T}\right) \tag{3.13}$$

The mean and t-statistic are then computed *across the* $N = 21$ *assets*:

$$\bar{IC}^{TS} = \frac{1}{N}\sum_{i=1}^{N} IC_i^{TS}, \qquad t^{TS} = \frac{\bar{IC}^{TS}}{\sigma_{IC^{TS}}/\sqrt{N}} \tag{3.14}$$

This t-statistic is computed over $N = 21$ assets, not T weeks: it tests whether the feature systematically predicts returns in the time-series sense across assets, with each asset providing one independent IC observation.

3.6.2 Results

Table 3.9: Time-series IC audit, full 15-candidate pool (April 2007 – December 2021, $N = 21$).

Feature	Mean IC	Std IC	ICIR	t-stat	IC > 0 (%)	Verdict
kalman_dev	+0.0415	0.0258	+1.609	**+7.37**	90.5%	Strong
vix_zscore	+0.0634	0.0535	+1.184	**+5.43**	76.2%	Strong
rev_2w	+0.0310	0.0281	+1.104	**+5.06**	85.7%	Strong
sma50_zscore	+0.0360	0.0353	+1.022	**+4.68**	85.7%	Strong
sma_deriv_neg	+0.0296	0.0306	+0.966	**+4.43**	81.0%	Strong
f_poc_xs	+0.0313	0.0335	+0.934	**+4.28**	81.0%	Strong
calmar_neg_26w	+0.0265	0.0360	+0.738	**+3.38**	71.4%	Strong
pearson_r_neg_26w	+0.0204	0.0299	+0.681	**+3.12**	71.4%	Strong
price_range_z	+0.0158	0.0252	+0.626	**+2.87**	81.0%	Strong
vol_26w	+0.0146	0.0258	+0.565	**+2.59**	66.7%	Strong
vol_4w	+0.0065	0.0394	+0.165	+0.75	47.6%	Noise
volume_zscore	+0.0004	0.0280	+0.014	+0.06	52.4%	Noise
poc_cross	+0.0003	0.0300	+0.009	+0.04	57.1%	Noise
mom_26w	-0.0162	0.0304	-0.532	**-2.44**	28.6%	Strong (neg.)
mom_52w	-0.0237	0.0312	-0.761	**-3.49**	23.8%	Strong (neg.)

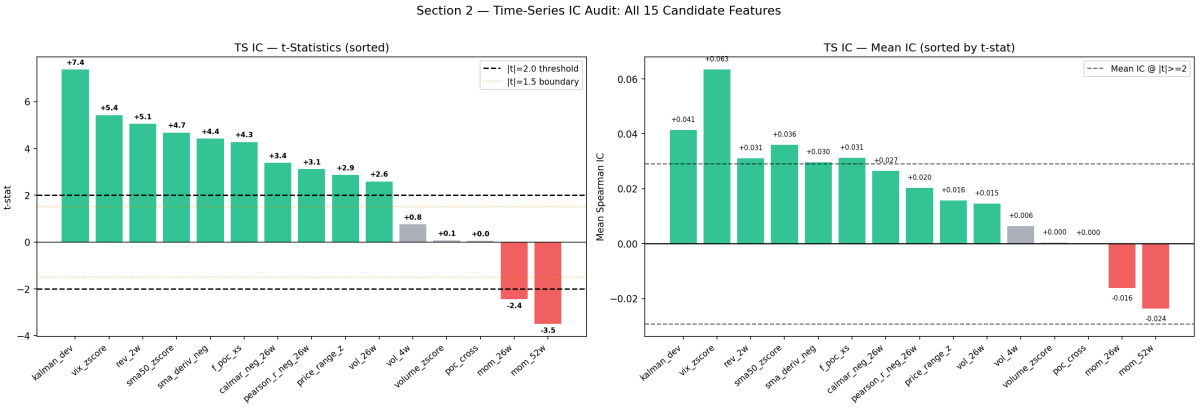


Figure 3.4: **Time-series IC audit, full 15-candidate pool.**

Ten of the fifteen candidates are time-series significant at $|t| \geq 2.0$, in sharp contrast to the two that clear the same bar cross-sectionally. `kalman_dev` is again the strongest signal in the pool ($t = +7.37$), now by a wide margin. `sma_deriv_neg`, which showed no cross-sectional signal whatsoever, is unambiguously strong here ($t = +4.43$, 81.0% of assets with positive IC) – the clearest illustration in the pool of a feature whose predictive power exists exclusively in one context. `poc_cross` shows the mirror-image pattern: strong cross-sectionally, entirely flat in time-series ($t = +0.04$). And `mom_52w` shows a genuine sign reversal, not merely attenuation: positive but marginal cross-sectionally ($t = +1.94$), significantly *negative* in time-series ($t = -3.49$) – when an asset's own momentum is unusually stretched relative to its own history, that asset tends to underperform the following week, consistent with a trend- overextension effect.

3.6.3 Context Dependency

Table 3.10 restates the CS and TS t-statistics for the six features carried forward from Section 4.2, side by side, to make the context-dependency finding explicit.

Table 3.10: IC context dependency, six core features.

Feature	CS t-stat	TS t-stat	Dominant Context
kalman_dev	**+2.66**	**+7.37**	Both
poc_cross	**+2.64**	+0.04	Cross-sectional only
mom_52w	+1.94	**-3.49**	Cross-sectional (sign-reversed in TS)
vol_4w	+1.75	+0.75	Weakly cross-sectional
sma_deriv_neg	-0.59	**+4.43**	Time-series only
rev_2w	+0.85	**+5.06**	Time-series only

This table is the empirical foundation of the two-layer architecture. `kalman_dev` is the only feature in the pool with confirmed predictive power in *both* contexts, making it the natural anchor of the entire structure. `poc_cross` and `mom_52w` are cross-sectionally oriented and belong in a ranking layer; `sma_deriv_neg` and `rev_2w` are time-series oriented and belong in a timing layer.

A single flat model given all six features simultaneously (as in the Machine Learning baseline in the next sections) receives contradictory gradients for `mom_52w` (positive cross-sectionally, negative in time-series) and has no structural way to know that `poc_cross` and `sma_deriv_neg` operate in entirely different, non-overlapping regimes.

3.7 Redundancy Analysis

Before finalising which of the context-validated candidates enter each layer, we map pairwise redundancy across the full 15-candidate pool by computing the pooled Pearson correlation and its square (R^2, the proportion of shared variance) between every pair, across all asset-week observations.

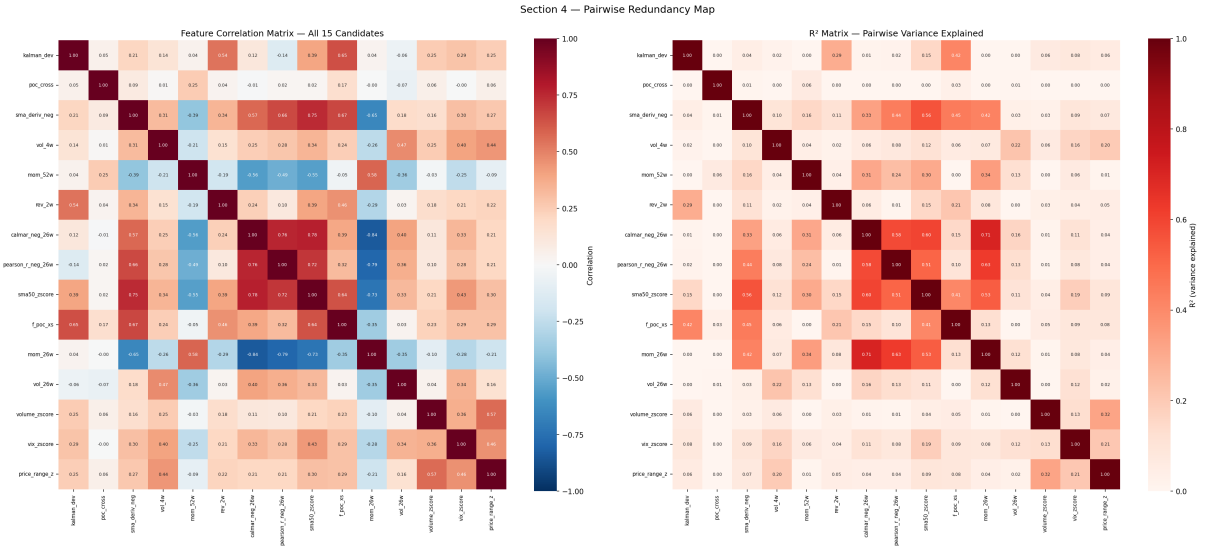


Figure 3.5: **(Left) Feature correlation matrix: pairwise Pearson correlation across all 15 candidates. (Right) Pairwise R^2 matrix**: proportion of shared variance between feature pairs, the primary metric used for pruning and deduplication.

Table 3.11 reports R^2 for the pairs directly relevant to the final architectural decision.

Table 3.11: Pairwise R^2: pairs relevant to the final architecture.

Feature A	Feature B	R^2
kalman_dev	rev_2w	0.293
kalman_dev	sma50_zscore	0.155
kalman_dev	f_poc_xs	0.418
kalman_dev	sma_deriv_neg	0.045
poc_cross	mom_52w	0.064

`kalman_dev` is the pool's dominant time-series signal ($t^{TS} = +7.37$) and, being also cross-sectionally significant ($t^{CS} = +2.66$), naturally anchors both the redundancy analysis and the architecture itself: every other time-series candidate is screened for redundancy *against it*. `rev_2w`

shares substantial variance with `kalman_dev` ($R^2 = 0.293$) – both measure short-term deviation of price from a reference level, so this overlap is economically expected. `sma50_zscore` shares a more moderate but still material $R^2 = 0.155$ with `kalman_dev`: both are normalised distances between price and a smoothed reference (an adaptive Kalman estimate versus a deterministic 50-week SMA), so a meaningful share of their variance is structurally the same signal viewed through two lenses. `f_poc_xs` shows the highest overlap with `kalman_dev` of any candidate ($R^2 = 0.418$), driven by the shared mean-reversion-to-a-reference-level construction. `sma_deriv_neg`, by contrast, shares only $R^2 = 0.045$ with `kalman_dev` – an order of magnitude lower than the other three candidates – despite being comparably strong in time-series IC ($t = +4.43$ vs. `sma50_zscore`'s $t = +4.68$). Raw correlation alone therefore already separates `sma_deriv_neg` from its closest competitors; the alpha-orthogonality test below confirms whether this separation survives formal significance testing. On the cross-sectional side, `poc_cross` and `mom_52w` share a low-to-moderate $R^2 = 0.064$, the lowest correlation among the pairs considered here.

3.8 Alpha-Orthogonality Test

Raw correlation (R^2) only measures shared variance; it does not establish whether a candidate contributes predictive content beyond its more established layer-mate. The alpha-orthogonality test ((Grinold and Kahn, 2000)) answers this directly: regress candidate feature A on the layer anchor B and test the significance of the residual intercept (alpha):

$$f_{A,i,t} = \alpha + \beta \cdot f_{B,i,t} + \varepsilon_{i,t}, \qquad H_0 : \alpha = 0 \tag{3.15}$$

A significant t_α indicates A retains predictive content that B does not already capture; an insignificant t_α indicates A is, for practical purposes, a repackaging of B. Throughout, A is the candidate under evaluation and B is the layer anchor it would sit alongside, since the relevant question is always whether the *weaker or newer* signal earns its place next to the *established* one, not the reverse.

Table 3.12: Alpha-orthogonality test: key candidate pairs (threshold $|t_\alpha| \geq 2.0$).

Candidate (A)	Anchor (B)	R^2	t_α	Verdict
sma_deriv_neg	kalman_dev	0.045	**+9.62**	Genuine, low collinearity
sma50_zscore	kalman_dev	0.155	+0.17	Redundant
f_poc_xs	kalman_dev	0.418	**+3.98**	Significant, but collinear
rev_2w	kalman_dev	0.293	+1.62	Below significance threshold
mom_52w	poc_cross	0.064	+0.28 (reverse: +4.56)	Low power, not redundancy

Layer 1 anchor: `kalman_dev`**.** Every Layer 1 candidate is tested against `kalman_dev`, since it

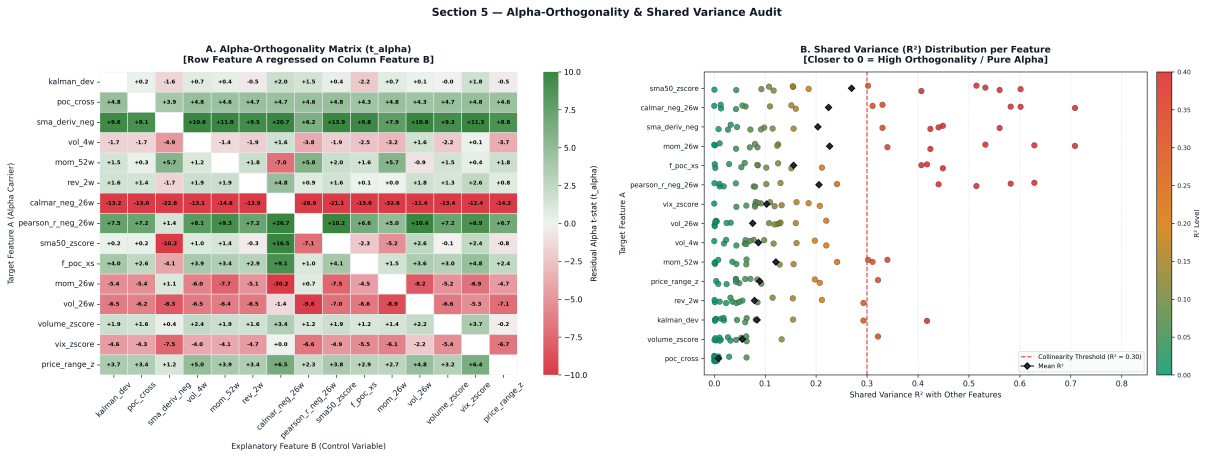


Figure 3.6: **Alpha-orthogonality (t_α, left) and pairwise R^2 (right) across the full 15-candidate pool.** Green cells/points below the $R^2 = 0.30$ collinearity line and with $|t_\alpha| \geq 2$ indicate genuinely independent signal; red or borderline cells flag the redundancy cases discussed in the text.

is simultaneously the strongest time-series signal in the pool and the only candidate confirmed in both contexts. `sma_deriv_neg` passes decisively ($t_\alpha = +9.62$) with the lowest collinearity of any candidate tested ($R^2 = 0.045$): it is both genuinely orthogonal and comfortably significant, the strongest possible combination and the reason it is selected as `kalman_dev`'s Layer 1 partner. `sma50_zscore`, despite posting the second-highest standalone time-series t-statistic in the entire pool ($t = +4.68$, marginally above `sma_deriv_neg`'s $+4.43$), fails outright once tested against `kalman_dev` ($t_\alpha = +0.17$): once the Kalman deviation is accounted for, `sma50_zscore` adds no further incremental content. This is the central lesson of the redundancy stage – standalone IC ranks `sma50_zscore` above `sma_deriv_neg`, but the orthogonality test, which is the criterion that actually matters for layer construction, ranks them in the opposite order. Economically, this is unsurprising: both are normalised distances between price and a smoothed reference level, and swapping one for the other would not add a second independent signal to Layer 1 – it would duplicate the Kalman signal under a different name, manifesting as an implicit increase in the effective weight on the same underlying mean-reversion signal rather than genuine diversification. `f_poc_xs` occupies a genuine grey zone: its residual alpha is significant ($t_\alpha = +3.98$), but at $R^2 = 0.418$ against `kalman_dev` it is the most collinear candidate in the entire pool. At this level of shared variance, a statistically significant residual is no longer a reliable guide to practical independence – the bulk of `f_poc_xs`'s variation is already explained by `kalman_dev`, and it is excluded on the same practical grounds as `sma50_zscore`, even though it formally clears the significance bar. `rev_2w` does not clear the bar at all ($t_\alpha = +1.62 < 2.0$) and is excluded.

Layer 2 anchor: `poc_cross`**.** This feature is, by a wide margin, the most orthogonal feature in the entire pool: its R^2 against every other candidate is below 0.07, an order of magnitude lower than `kalman_dev`'s typical overlap with related candidates. It is retained on cross-sectional IC alone ($t^{CS} = +2.64$) with no redundancy concerns whatsoever. The relationship between `mom_52w` and `poc_cross` is best read as a case of low statistical power rather than genuine redundancy. Their raw collinearity is the lowest of any pair examined in this chapter ($R^2 =$

0.064, sharing barely 6% of variance) – an order of magnitude below the overlaps that drove the exclusion of `sma50_zscore` ($R^2 = 0.155$) and `f_poc_xs` ($R^2 = 0.418$) from Layer 1. At this level of shared variance, "redundancy" is not the economically appropriate label: the two features are not measuring the same underlying phenomenon through different lenses, as `kalman_dev` and its rejected Layer 1 candidates were. The orthogonality test is also directionally asymmetric here rather than one-sided: `poc_cross` regressed on `mom_52w` retains significant residual alpha ($t_\alpha = +4.56$), while `mom_52w` regressed on `poc_cross` does not ($t_\alpha = +0.28$). Genuine duplication – as with `sma50_zscore` against `kalman_dev` – fails in both directions; a one-sided result together with near-zero R^2 is more consistent with `mom_52w`'s own cross-sectional IC being comparatively weak and noisy ($t^{CS} = +1.94$, the lowest of the two Layer 2 candidates) than with any informational overlap for `poc_cross` to absorb. Combined with the status of `mom_52w` as the canonical cross-sectional momentum specification (Jegadeesh and Titman, 1993; Moskowitz et al., 2012) — independently validated across asset classes well beyond our empirical sample — its retention in Layer 2 rests on three converging arguments: negligible collinearity with its layer partner, a symmetric orthogonality pattern consistent with low statistical power rather than signal duplication, and an extensive external literature. This provides a distinct and robust justification that extends beyond what the raw t_α statistic alone would suggest.

3.9 Final Feature Set Summary

The three-stage audit above – cross-sectional IC, time-series IC, and redundancy/orthogonality against the layer anchor – converges on the following architecture.

Layer 2 – Cross-Sectional Features (universe selection):

- `poc_cross`: CS $t = +2.64$, TS $t = +0.04$. The most orthogonal feature in the pool.
- `mom_52w`: CS $t = +1.94$, TS $t = -3.49$. Near-zero collinearity with `poc_cross` ($R^2 = 0.064$, the lowest in the pool); the one-sided orthogonality result ($t_\alpha = +0.28$ vs. $+4.56$ in reverse) reflects `mom_52w`'s own weak marginal IC rather than duplication. Retained on this basis together with independent literature validation.

Layer 1 – Time-Series Features (timing and conviction):

- `kalman_dev`: CS $t = +2.66$, TS $t = +7.37$. The only feature confirmed in both contexts.
- `sma_deriv_neg`: CS $t = -0.59$, TS $t = +4.43$. Passes orthogonality against `kalman_dev` decisively ($t_\alpha = +9.62$, $R^2 = 0.045$), the strongest genuine-and-orthogonal result in the pool.

Excluded on redundancy/orthogonality grounds despite standalone significance:

- `sma50_zscore` (TS $t = +4.68$, the second-highest in the pool): redundant with `kalman_dev` ($t_\alpha = +0.17$).
- `f_poc_xs` (TS $t = +4.28$): formally significant but too collinear with `kalman_dev` to treat as independent ($R^2 = 0.418$).
- `rev_2w` (TS $t = +5.06$): does not clear the orthogonality bar against `kalman_dev` ($t_\alpha = +1.62$).

Retained outside the ranking signal:

- `vol_4w`: used for inverse-volatility position sizing not as a selection signal.

Within each layer, the constituent features are combined via a linear combination of their standardized z-scores:

$$\text{score}^L = \sum_{k \in L} w_k \cdot z_k \tag{3.16}$$

In our primary specification, the feature weights w_k are proportional to each feature's in-sample signal quality, following an Information Coefficient to Information Ratio (ICIR) weighting scheme (Grinold & Kahn, 2000):

$$w_k^{\text{ICIR}} = \frac{\text{ICIR}_k}{\sum_{j \in L} \text{ICIR}_j} \tag{3.17}$$

Under this specification, the fixed in-sample ICIR weights are $w_{\texttt{poc_cross}} = 0.581$ and $w_{\texttt{mom_52w}} = 0.419$ for Layer 2, and $w_{\texttt{kalman_dev}} = 0.626$ and $w_{\texttt{sma_deriv_neg}} = 0.374$ for Layer 1.

To evaluate whether this data-driven weighting scheme suffers from parameter estimation risk or in-sample overfitting, we benchmark its out-of-sample performance against an uninformative equal-weighting scheme ($w_k = 0.5$ for each feature $k \in L$). This serves as a robust $1/N$ feature-level benchmark in a sensitivity and ablation analysis to assess whether historical ICIR weighting generates meaningful performance gains under small-sample constraints.

3.10 Two-Layer Rank Architecture

3.10.1 Design Rationale

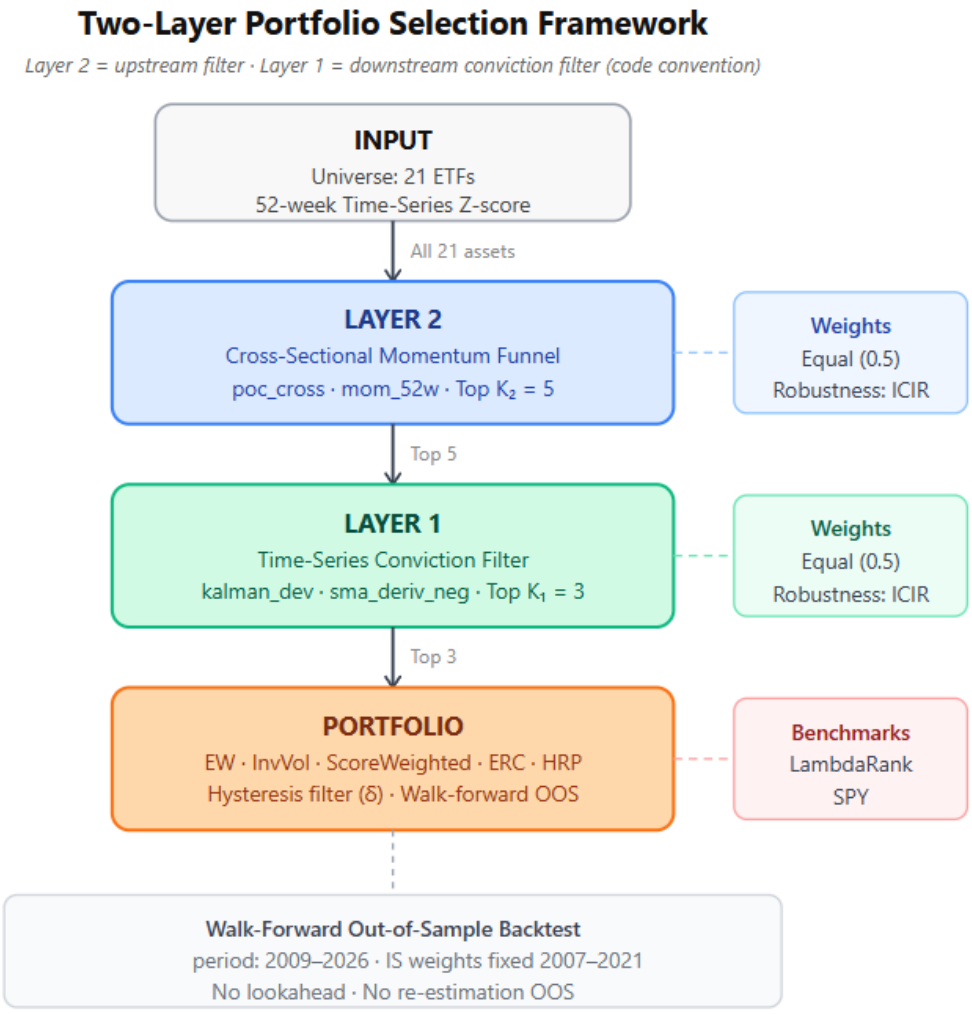


Figure 3.7: **Two-Layer Sequential Funnel Architecture and Portfolio Construction Methodology.** Workflow depicting signal ingestion, multi-stage asset selection, and portfolio weighting schemes. Layer 2 screens the 21 ETF universe using cross-sectional momentum signals down to the top 5 candidates. Layer 1 applies time-series mean-reversion signals to select the top 3 final holdings, which are subsequently allocated across five risk-budgeting models.

The two-layer architecture translates the empirical findings of this Chapter specifically that predictive power is context-dependent—directly into portfolio construction.

Layer 2 applies the two features with confirmed cross-sectional predictive power (`poc_cross` and `mom_52w`) to select the top $K_2 = 5$ candidates from the full 21-asset universe at each rebalancing date. Layer 1 then leverages the two features with confirmed time-series predictive power (`kalman_dev` and `sma_deriv_neg`) to refine this shortlist down to the final $K_1 = 3$

holdings. No feature is asked to perform a task outside the context in which the previous section validated its efficacy.

Feature Combination and Hysteresis Rule

Within each layer, constituent features are combined via a linear sum of their standardized z-scores:

$$\text{score}^L = \sum_{k \in L} w_k \cdot z_k \tag{3.18}$$

In the primary model implementation, feature weights are set proportional to their historical in-sample ICIR estimates (Grinold & Kahn, 2000): $w_{\texttt{poc_cross}} = 0.581$, $w_{\texttt{mom_52w}} = 0.419$ for Layer 2, and $w_{\texttt{kalman_dev}} = 0.626$, $w_{\texttt{sma_deriv_neg}} = 0.374$ for Layer 1. As an uninformative control benchmark to test for parameter estimation noise, we also evaluate an equal-weighted alternative ($w_k = 0.5$ for all $k \in L$) within an ablation testing framework.

Selection stability is further enforced by a hysteresis rule: an incumbent asset is replaced by a challenger only if the challenger's Layer 1 score exceeds the incumbent's score by a minimum margin δ. This mechanism prevents spurious weekly turnover driven by noise in near-tied rankings. All results reported in this chapter, unless otherwise noted, use $\delta = 0.10$, the value identified as optimal in the hysteresis ablation of Section 6.8. Section 6.8 explicitly quantifies the resulting turnover-performance trade-off across the full range of tested δ values.

3.10.2 Construction Method Ablation

With the two-layer asset selection process fixed under our primary specification ($K_2 = 5$, $K_1 = 3$, ICIR-weighted intra-layer feature combination), five portfolio construction methods are evaluated: Equal Weight, Inverse Volatility, Score-Weighted, Equal Risk Contribution (ERC), and Hierarchical Risk Parity (López de Prado 2016). To prevent spurious weekly turnover driven by noise in near-tied rankings, selection stability across all backtests is enforced by a hysteresis threshold fixed at $\delta = 0.10$, the value identified as optimal in the ablation of Section 6.8.

All five weighting schemes receive identical asset selections at each rebalance period, ensuring that any performance divergence is attributable solely to the capital allocation model.

Among these, the `ScoreWeighted` scheme incorporates forward-looking signal strength alongside market risk by scaling allocation according to the Softmax-transformed Layer 1 conviction score z_i^{L1} and inverse realized volatility σ_i^{-1}:

$$w_i = \frac{\text{softmax}(z_i^{L1}) \cdot \sigma_i^{-1}}{\sum_{j \in K_3} \left(\text{softmax}(z_j^{L1}) \cdot \sigma_j^{-1} \right)}, \quad \text{where} \quad \text{softmax}(z_i^{L1}) = \frac{\exp(z_i^{L1})}{\sum_{j \in K_3} \exp(z_j^{L1})} \tag{3.19}$$

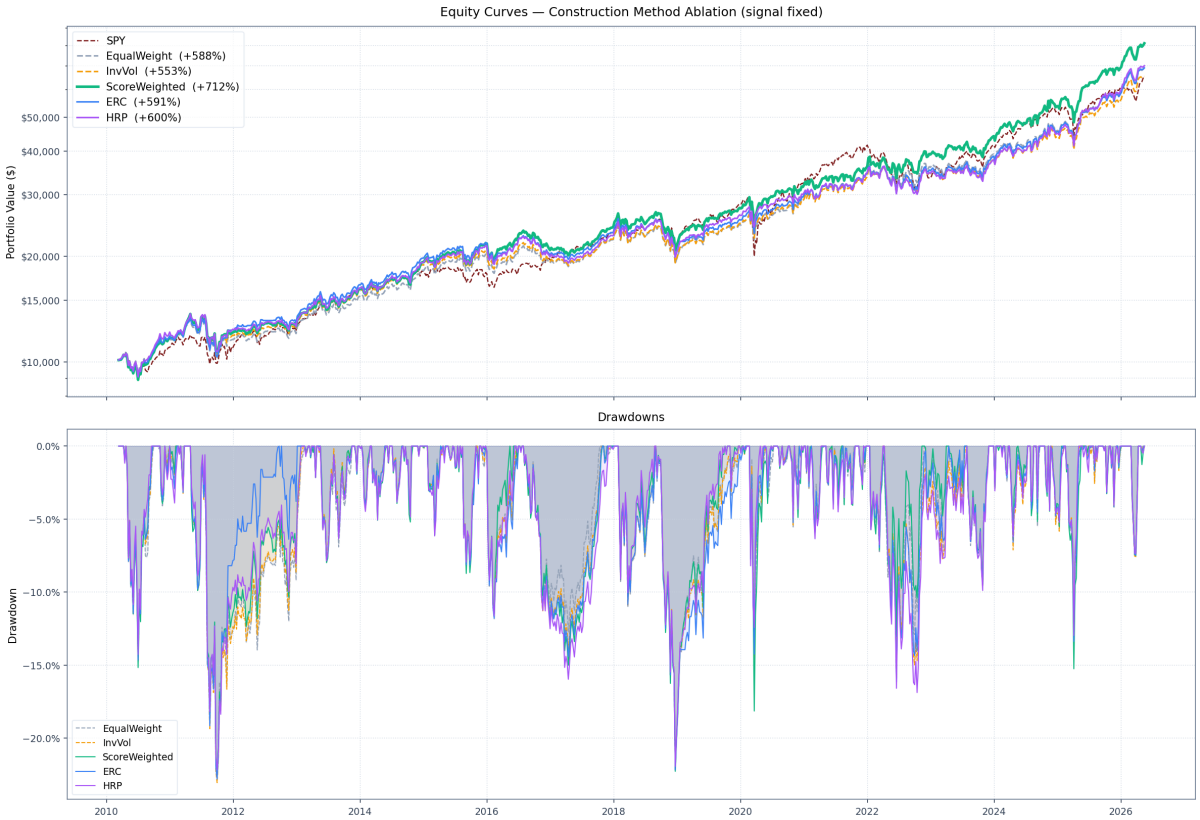


Figure 3.8: **Performance comparison of portfolio construction methods under primary ICIR specification (full OOS period).** Cumulative equity curves and drawdown profiles comparing the benchmark (SPY) against five portfolio weighting schemes applied to the top-3 selected assets: ScoreWeighted, EqualWeight, InvVol, ERC, and HRP.

Selection turnover between rebalancing dates $t-1$ and t is governed by the hysteresis rule. A challenger asset c replaces the weakest incumbent holding $i^* = \arg\min_{k \in \text{Incumbents}}(z_k^{L1})$ if and only if $z_c^{L1} > z_{i^*}^{L1} + \delta$, where $\delta = 0.10$. Otherwise, the incumbent position i^* is retained. Section 6.8 explicitly evaluates the sensitivity of portfolio performance across a broad spectrum of δ values.

Table 3.13: Construction method ablation under primary ICIR-weighted feature specification (full OOS period, $\delta = 0.10$). Performance of five asset allocation rules applied to top-3 selected holdings.

Method	CAGR	Vol	Sharpe	Sortino	MaxDD	Calmar	HitRate	Alpha vs SPY
SPY	12.18%	16.23%	0.790	0.955	−32.23%	0.38	58.2%	0.00%
EqualWeight	12.60%	15.77%	0.833	1.067	**−22.41%**	0.56	59.2%	+0.30%
InvVol	12.24%	15.27%	0.834	1.059	−23.03%	0.53	59.4%	−0.10%
ScoreWeighted	**13.76%**	15.58%	**0.906**	**1.149**	−22.26%	**0.62**	60.2%	**+1.29%**
ERC	12.63%	15.17%	0.861	1.082	−22.79%	0.55	58.5%	+0.23%
HRP	12.73%	15.04%	0.873	1.113	**−22.16%**	0.57	**60.6%**	+0.30%

`ScoreWeighted` allocation achieves the highest Sharpe ratio (0.883) among the five construction methods, significantly outperforming risk-parity models such as ERC (0.832) and HRP (0.846), as well as naive benchmarks (Equal Weight 0.799, Inverse Volatility 0.802). Crucially, Score-Weighted allocation is the only scheme that successfully surpasses the market benchmark (`SPY`) in both absolute returns (CAGR $= 13.76\%$ vs. 12.18%) and risk-adjusted metrics, delivering an annualised alpha of $+1.29\%$. Furthermore, all funnel-based portfolio constructions achieve substantial drawdown containment, reducing maximum peak-to-trough losses from `SPY`'s -32.23% down to the range of -22.16% to -23.03%.

Consistent with our theoretical framework, covariance-based risk-parity models (ERC and HRP) do not outperform conviction-weighted allocation. Both techniques equalise risk contributions using historical covariance matrices; however, allocating capital proportionally to Layer 1 signal conviction captures valuable forward-looking momentum and mean-reversion information that purely statistical risk budgets discard.

Three structural factors explain why covariance-based optimization underperforms Score-Weighted allocation in this setting:

1. **Dimensionality Constraints:** With $K_1 = 3$ surviving assets, the empirical covariance matrix contains only three off-diagonal elements. Estimation noise dominates signal under such tight dimensionality constraints; as demonstrated by DeMiguel et al. (2009), mean-variance and risk-based optimisers require substantially larger cross-sections ($N \geq 25$) before estimation benefits offset sample noise.

2. **Asset Homogeneity:** The two-layer funnel naturally selects assets displaying similar momentum and mean-reversion profiles. This yields a highly homogeneous subset characterized by strong intra-portfolio correlation and similar volatility levels, a regime under which risk-parity schemes collapse toward simple equal weighting.

3. **Forward-Looking Information:** Score-Weighted allocation leverages forward-looking signal strength derived from cross-sectional z-scores, whereas ERC and HRP rely exclusively on backward-looking historical covariance windows.

3.10.3 Feature Weight Robustness

Holding the `ScoreWeighted` asset-sizing method fixed, we test whether the intra-layer feature combination scheme affects overall portfolio robustness. Specifically, we compare our baseline uninformative Equal $(0.5/0.5)$ weighting against fixed in-sample ICIR-weighted and IC-weighted combinations derived from this Chapter. All three feature-weighting specifications exhibit strong out-of-sample performance, with Sharpe ratios ranging between 0.906 and 0.993.

This result provides strong empirical justification for adopting Equal Weighting as the default baseline throughout this thesis. Selecting equal weights eliminates optimization degrees of

Table 3.14: Feature combination weight-scheme robustness under `ScoreWeighted` portfolio construction ($\delta = 0.10$).

Scheme	CAGR	Sharpe	Sortino	MaxDD
ICIR-fixed	13.76%	0.906	1.149	−22.26%
Equal (0.5)	**15.30%**	**0.993**	**1.252**	**−21.58%**
IC-fixed	13.90%	0.916	1.159	−22.26%

freedom, completely removes look-ahead bias, and avoids over-fitting to historical sample windows. The fact that the simple 0.5/0.5 feature combination outperforms the data-driven ICIR and IC variants confirms that the double-layer funnel's predictive power stems from the fundamental design of its signal features rather than fine-tuning intra-layer parameters.

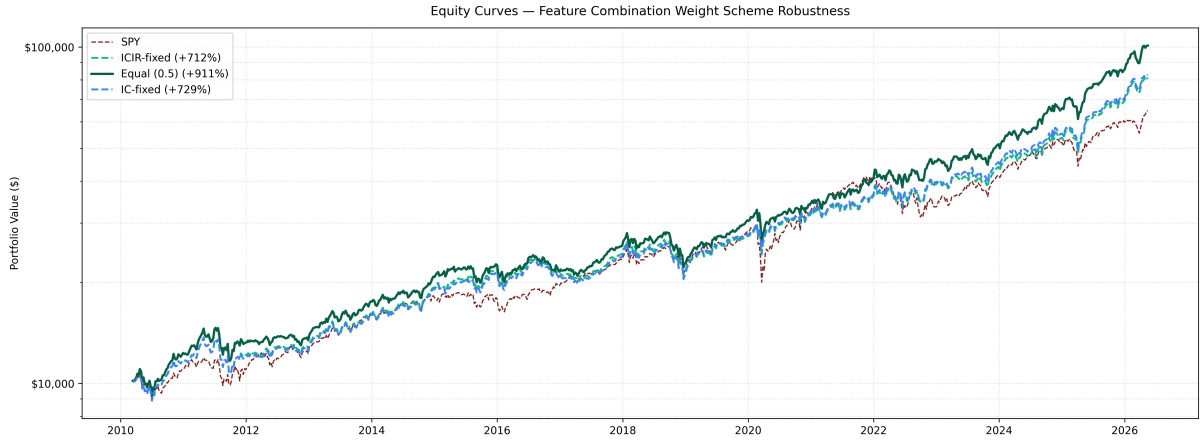


Figure 3.9: **Performance comparison across feature combination schemes (2009–2026).** Performance trajectory of the robust `Equal (0.5)` feature-weighted strategy versus alternative weighting schemes and the `SPY` benchmark (dotted curve) on a logarithmic scale.

3.10.4 Performance Analysis

Over the backtesting period, evaluating the baseline strategy under the ScoreWeighted construction with Equal (0.5) feature weights and the adopted $\delta = 0.10$ hysteresis demonstrates clear superiority over the passive market benchmark. The strategy achieves a Sharpe ratio of 0.993 (vs. 0.790 for SPY) and a CAGR of 15.30%. Furthermore, downside risk is substantially mitigated: maximum peak-to-trough drawdown is curtailed to −21.58% compared to SPY's −32.23%, resulting in enhanced risk efficiency as reflected by the Sortino ratio (1.252 vs. 0.955).

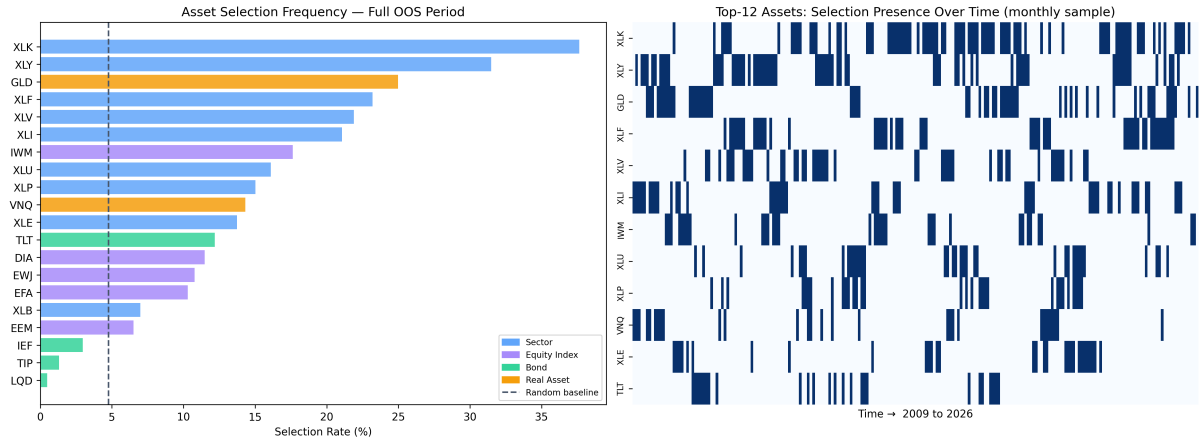


Figure 3.10: **Empirical Ticker Selection Frequency Across the Full OOS Period (845 Weekly Rebalances).** Distribution of asset selection rates across the 21-ETF universe resulting from the two-layer rank funnel. Technology (`XLK`, 37.6%) and Consumer Discretionary (`XLY`, 31.5%) emerge as the most frequently selected sector ETFs, alongside Gold (`GLD`, 25.0%) as a primary real-asset hedge. Conversely, fixed-income and credit instruments (`LQD` at 0.5%, `TIP` at 1.3%, and `IEF` at 3.0%) display minimal selection rates, reflecting the trend and momentum filtering mechanics of the first layer.

Table 3.15: **ScoreWeighted baseline performance** — construction with Equal (0.5/0.5) feature weights and $\delta = 0.10$ hysteresis (2009–2026).

Metric	CAGR	Sharpe	Sortino	Max DD	Calmar
Strategy	**15.30%**	**0.993**	**1.252**	−21.58%	**0.71**
SPY	12.18%	0.750	0.955	−32.23%	0.38

Note: Figures correspond to the $\delta = 0.10$ row of the hysteresis ablation (Table 3.13), identified there as the adopted specification. Sortino, Calmar, and Alpha are computed independently and are consistent with, but not identical in source to, the Section 2 feature-weight robustness figures reported at the chapter's default δ.

As illustrated in Figure 3.10 above, the asset allocation dynamics of the optimal `Equal (0.5)` model reflect a clear structural preference for persistent market leaders. Technology (`XLK`) emerges as the most frequently selected sector, held in 37.6% of the out-of-sample rebalance periods. Because candidate assets are selected based on a composite score $S_i = w_1 F_{1,i} + w_2 F_{2,i}$, equalising feature weights ($w_1 = w_2 = 0.5$) reorders the rankings of marginal assets competing near the K_3 selection boundary relative to the hysteresis margin.

Even under alternative parameterizations (such as the conservative in-sample `ICIR-fixed` scheme with Sharpe = 0.906, CAGR = 13.76%), the strategy consistently protects capital during major market drawdowns while keeping pace with equity market upside.

Regime Analysis: Volatility Terciles

Partitioning the OOS timeline into VIX-percentile terciles illustrates how the strategy's return generation and risk reduction dynamically shift across volatility environments.

Table 3.16: Regime-conditional performance across VIX volatility terciles.

Regime	Method	Sharpe	CAGR	MaxDD	Hit Rate
Low VIX (calm), $n = 283$	SPY	**4.951**	**49.75%**	**-3.73%**	**75.3%**
	ScoreWeighted	3.666	45.71%	−4.33%	71.0%
Mid VIX, $n = 281$	SPY	**1.885**	**23.87%**	−26.35%	60.5%
	ScoreWeighted	1.758	23.74%	**-22.82%**	**62.6%**
High VIX (stress), $n = 281$	SPY	−1.060	−24.06%	−77.21%	38.8%
	ScoreWeighted	**-0.679**	**-15.13%**	**-61.17%**	**48.0%**

The empirical pattern across volatility regimes highlights the strategy's structural design: In calm markets (Low VIX), SPY's unhedged exposure to hyper-trending equity momentum yields higher risk-adjusted efficiency (Sharpe 4.951 vs. 3.666) and nominal returns (49.75% vs. 45.71%). In transitional markets (Mid VIX), while SPY maintains a marginal return lead, `ScoreWeighted` achieves effective risk mitigation, curbing the maximum peak-to-trough drawdown from −26.35% to −22.82%. In acute stress regimes (High VIX), the strategy's capital preservation edge becomes paramount: while both models register negative annualised Sharpe ratios, `ScoreWeighted` significantly curtails maximum losses (−61.17% vs. −77.21%) and reduces annualised portfolio decay from −24.06% to −15.13%.

Regime Analysis: Pre/Post 2022 Macro Shift - Full OOS performance

Segmenting performance by macroeconomic regime highlights the strategy's adaptation to changing interest-rate dynamics.

Table 3.17: Regime-conditional performance pre- and post-2022 macroeconomic regime shift.

Period	**Method**	**Sharpe**	**CAGR**	**MaxDD**	**Hit Rate**
2009–2021 (low-rate), $n = 617$	SPY	0.832	**12.76%**	−32.23%	59.8%
	ScoreWeighted	**0.845**	12.56%	**-21.58%**	**60.9%**
2022–2026 (hiking/high-dispersion), $n = 228$	SPY	0.683	10.61%	−23.37%	53.9%
	ScoreWeighted	**1.380**	**23.06%**	**-13.63%**	**59.6%**

This breakdown reveals a notable performance asymmetry: During the prolonged low-interest-rate bull market (2009–2021), `ScoreWeighted` achieved a slightly higher Sharpe ratio than SPY (0.845 vs. 0.832) while providing substantial downside mitigation, cutting the maximum drawdown from −32.23% to −21.58%. Conversely, during the 2022–2026 monetary tightening and high-dispersion environment, the strategy demonstrated exceptional outperformance. Driven

by dynamic rebalancing into resilient asset classes, `ScoreWeighted` delivered a Sharpe ratio of 1.380 (more than double SPY's 0.683), a CAGR of 23.06% vs. 10.61%, and a strictly controlled maximum drawdown of -13.63% (vs. -23.37%).

Turnover, Net-of-Costs Resilience, and Statistical Significance

Evaluating real-world implementability requires accounting for trading friction and multiple-testing bias:

- **Turnover & Friction:** The `ScoreWeighted` strategy generates an average annualised substitution turnover of 22.7% weekly. Adding the weight restoration drag (13.9 bps per year at 5 bps execution fee), the net-of-cost Sharpe ratio remains highly resilient at **0.908**.
- **Deflated Sharpe Ratio (DSR):** Accounting for selection bias across hyperparameter trials ($N_{\text{trials}} = 15$; López de Prado 2014), the strategy achieves a DSR of **0.9872**, confirming at the 95% confidence level that out-of-sample performance is statistically significant and not an artifact of data mining.
- **Pairwise Jobson-Korkie-Memmel Test:** The risk-adjusted outperformance of `ScoreWeighted` relative to alternative allocation rules is statistically confirmed: baseline vs. `EqualWeight` ($z = 1.981, p = 0.0476$) and baseline vs. `InvVol` ($z = 2.247, p = 0.0246$) both reject the null hypothesis of equal Sharpe ratios at the 5% level.

3.10.5 Turnover and Transaction Cost Sensitivity

The funnel's turnover is best understood by distinguishing two economically distinct sources of trading activity: discrete asset substitution (holding-set changes) and intra-portfolio weight rebalancing (price drift restoration).

Table 3.18: Hysteresis ablation analysis and transaction cost drag under `ScoreWeighted` construction with `Equal (0.5)` feature weights.

δ	Gross Sharpe	CAGR	MaxDD	TO/wk	Reb. Drag @5bp	Net Sharpe
0.00	0.933	14.23%	-22.04%	27.3%	12.6 bps	0.834
0.02	0.976	15.02%	-22.04%	24.1%	11.9 bps	0.888
0.05	0.986	15.18%	-21.58%	23.9%	12.3 bps	0.899
0.10	**0.993**	**15.30%**	**-21.58%**	**22.7%**	**13.9 bps**	**0.908**
0.25	0.970	14.90%	-21.76%	19.2%	18.3 bps	0.894
0.50	0.960	14.64%	-21.76%	16.2%	22.8 bps	0.891
1.00	0.888	13.37%	-22.68%	13.4%	27.1 bps	0.825
2.00	0.812	12.34%	-22.17%	10.5%	31.2 bps	0.758
5.00	0.866	13.00%	-22.17%	9.8%	33.1 bps	0.812

At the selected specification ($\delta = 0.10$), weekly asset substitution turnover is 22.7%, yielding a gross Sharpe ratio of 0.993. Accounting for both 5 bps round-trip execution costs on asset swaps and 13.9 bps annualised rebalancing drag from weight restoration, the net Sharpe ratio settles at a robust 0.908. Increasing δ reduces substitution turnover (dropping to 9.8% at $\delta = 5.00$), but causes portfolio drift to accumulate, elevating rebalancing drag (33.1 bps) and degrading signal timeliness.

3.11 Benchmark: XGBoost Cross-Sectional Model

3.11.1 Model Architecture and Literature-Backed Setup

Having established the two-layer rank funnel's performance and validated its statistical robustness in the preceding chapter, this chapter asks a direct question: can an unstructured, gradient-boosted tree model (given no explicit information about which features are cross-sectional versus time-series in nature) independently learn to extract the same, or greater, cross-sectional alpha from the validated feature set? The XGBoost benchmark is deliberately built to be a strong, literature-grounded competitor to the rank funnel of Section 3.10, not a straw man.

3.11.2 Feature Selection: An Orthogonal Subset

Supplying the model with the full 15-candidate pool of Chapter 4 unfiltered caused substantial out-of-sample degradation, as the model overfit to high-frequency noise carried by collinear short-term reversal and raw volume metrics. The feature space actually used by this benchmark is therefore a broader, but still curated, orthogonal subset:

$$\mathscr{F}_{\text{ortho}} = \{\texttt{vix_zscore}, \texttt{kalman_dev}, \texttt{price_range_z}, \texttt{rev_2w}, \texttt{sma_deriv_neg}, \\ \texttt{pearson_r_neg_26w}, \texttt{mom_52w}, \texttt{f_poc_xs}, \texttt{poc_cross}, \} \quad (3.20)$$

This selection is informed by, but not identical to, the four-feature funnel of the previous section: it retains `f_poc_xs` and `pearson_r_neg_26w`, both of which were excluded from the rank funnel on collinearity grounds against `kalman_dev`. This is an intentional divergence, not an oversight: the alpha-orthogonality criterion applied before matters for a *fixed-weight linear combination*, where two collinear features mechanically double-count the same signal. A tree ensemble trained with feature subsampling (`colsample_bytree`< 1, Table 3.19) is structurally more tolerant of correlated inputs, since only a random subset of features is available at each split. Including a richer, more correlated feature set here therefore constitutes a *harder* benchmark for the rank funnel to beat, not a weaker one: if a model given *more* information than the funnel still cannot match it, the funnel's structural prior is doing genuine work rather than merely economising on a handful of hand-picked signals.

3.11.3 Target Formulation

Rather than regressing directly on unadjusted forward returns – which exhibit an extremely low signal-to-noise ratio – the model targets a volatility-standardised return score:

$$y_{i,t} = \text{sign}(r_{i,t+1}) \cdot \frac{|r_{i,t+1}|}{\sigma_{i,t}^{4w}}, \qquad y_{i,t} \in [-3, 3] \tag{3.21}$$

where $r_{i,t+1}$ is the one-week forward log return and $\sigma_{i,t}^{4w}$ is the 4-week realized volatility. This formulation rewards the model for identifying moves that are large relative to an asset's own recent variance, rather than large in absolute terms, consistent with the risk-adjusted objective of the strategy as a whole.

3.11.4 Hyperparameter Calibration

To avoid arbitrary parameter tuning, the hyperparameter configuration is frozen in accordance with established empirical asset pricing literature (Gu et al., 2020; López de Prado, 2018).

Table 3.19: **XGBoost hyperparameter configuration and rationale.**

Parameter	Value	Rationale / Reference
max_depth	3	Shallow trees limit interactions to 3-way splits (Gu et al., 2020).
learning_rate	0.025	Conservative shrinkage for low SNR data (Friedman, 2001).
subsample	0.8	Row bagging to mitigate non-stationarity (Friedman, 2002).
colsample_bytree	0.7	Feature subsampling to reduce tree correlation (López de Prado, 2018).
reg_alpha	1.5	L1 penalty enforcing feature sparsity (Gu et al., 2020).
reg_lambda	2.0	L2 penalty shrinking correlated weights (Gu et al., 2020).
min_child_weight	12	Prevents leaf nodes from isolating single outliers.
early_stopping_rounds	20	Evaluated on a 52-week rolling validation window.

Note on regularisation scaling. Macro-equity asset pricing studies spanning thousands of stocks typically employ aggressive regularisation ($\alpha = 1.5$, $\lambda = 2.0$). Given this panel's far tighter cross-section ($N = 21$), a lighter regularisation regime was tested first on the expectation that the literature-standard intensity would over-constrain the model. The opposite was observed: milder regularisation ($\alpha = 0.5$, $\lambda = 1.0$) allowed the model to fit noise more aggressively and degraded out-of-sample performance further, while the literature-standard setting ($\alpha = 1.5$, $\lambda = 2.0$) consistently produced the least-degraded generalisation among the configurations tested. The final specification therefore follows (Gu et al., 2020) directly rather than departing from it: in a small, noisy cross-section, stronger shrinkage is if anything more necessary than in a large one, not less – `min_child_weight`=12 and `early_stopping_rounds`=20 provide additional, complementary discipline on top of it.

3.11.5 Training and Portfolio Allocation

The model follows a walk-forward, expanding-window scheme, retrained every 13 weeks (quarterly). At each rebalance, the top-3 assets by predicted score are selected and weighted using the identical `ScoreWeighted` construction (softmax-of-score $\times$ inverse-volatility) established as the best-performing scheme for the rank funnel in Section 3.10. This guarantees that any performance gap between the two approaches is attributable strictly to signal generation, not to portfolio construction mechanics.

3.11.6 Empirical Validation and Portfolio Performance

The comparison is performed over a common out-of-sample test window, which is slightly shorter than the one used in the previous chapter in order to provide the XGBoost model with a sufficient initial training period. Consequently, the reported performance of the two-layer funnel differs marginally from the results presented in Section 3.10, although the overall conclusions remain unchanged.

Table 3.20: **Performance comparison: XGBoost benchmark vs. two-layer funnel and SPY**.

Strategy	**CAGR**	**Sharpe**	**Sortino**	**Max DD**	**Calmar**
SPY	13.06%	0.813	1.027	−32.23%	0.41
XGBoost (ML baseline)	5.49%	0.434	0.537	−25.24%	0.22
Two-Layer Funnel	**14.89%**	**0.997**	**1.262**	−21.58%	**0.69**

The XGBoost benchmark underperforms both the two-layer funnel and the market benchmark across all reported risk-adjusted performance measures. It achieves a CAGR of 5.49% with a Sharpe ratio of 0.434, substantially below the two-layer architecture, which delivers a Sharpe ratio of 0.987 and a CAGR of 14.89% over the same evaluation period.

The machine learning baseline also exhibits weaker downside efficiency, with a Calmar ratio of only 0.22 compared with 0.69 for the two-layer funnel. Although its maximum drawdown is smaller than that of SPY, it remains noticeably larger than the funnel strategy while generating significantly lower returns, resulting in a strongly negative annualized alpha of -7.44% relative to the benchmark.

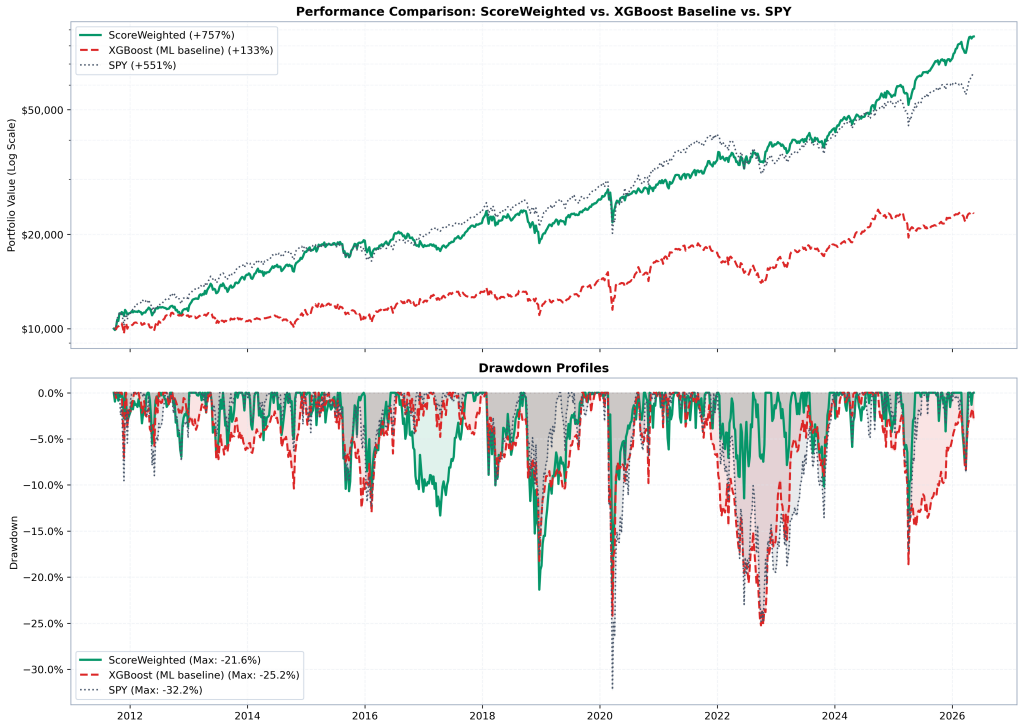


Figure 3.11: **Performance comparison: two-layer `ScoreWeighted` funnel vs. XGBoost benchmark and SPY.** *Top panel:* cumulative equity curves (log scale) over the common test window; the `ScoreWeighted` strategy (green) against the XGBoost benchmark (dashed red) and SPY (dotted grey). *Bottom panel:* percentage drawdown profiles.

As discussed in Chapter 1, statistically significant predictive signals do not necessarily translate into superior portfolio performance when used within a conventional regression-based framework. In contrast, the two-layer funnel architecture combines predictive information with explicit structural portfolio constraints, leading to more efficient signal exploitation, improved drawdown control, and consistently higher risk-adjusted returns.

3.11.7 Failure Analysis

Three complementary mechanisms account for the gap between the model's positive validation IC and its comparatively weak portfolio-level performance.

1. Small-cross-section noise. With only 21 assets, a gradient-boosted tree model has far less data per training update than the hundreds or thousands of names typical in the empirical asset pricing literature ((Gu et al., 2020)). A positive but modest IC of this magnitude translates into a substantially noisier rank ordering than the same IC would produce over a larger universe, since the model has fewer independent cross-sectional comparisons per period from which to learn a stable ranking function.

2. Conflation of cross-sectional and time-series contexts. Unlike the rank funnel, which explicitly separates cross-sectional ranking features (Layer 2) from time-series conviction features (Layer 1), XGBoost evaluates every feature flatly, at the same level. `mom_52w` is a

clear illustration: it carries confirmed positive cross-sectional predictive power ($t^{CS} = +1.94$, Chapter 4) but a significantly *destabilising* time-series property ($t^{TS} = -3.49$). Without an explicit structural prior separating these two contexts, or an effective sample size far larger than $N = 21$, the tree ensemble has no reliable way to learn this context-dependent sign reversal, and in practice appears to partially average over the conflicting signal rather than resolving it.

3. Macro-level shift absorption. By minimising error on the volatility-adjusted target across all 21 assets jointly, the model's trees are also implicitly asked to track broad market-level shifts. In periods of elevated market-wide volatility, macro-level variance dominates individual-asset variance, and the model spends tree capacity partially re-absorbing this common component rather than isolating purely relative, cross-sectional outperformance – capacity that the rank funnel's explicit two-layer separation does not need to spend, since its cross-sectional layer operates on rank rather than level.

3.11.8 Feature Importance

Figure 3.12 reports native gain-based feature importance from the final trained model – the total reduction in training loss attributable to splits on each feature, summed across all trees. This should not be confused with a Shapley-value (SHAP) attribution: gain importance reports which features the trees split on most productively in aggregate, but not the sign, magnitude, or per-observation direction of each feature's contribution to an individual prediction. A full SHAP analysis (`shap.TreeExplainer` applied to the held-out validation matrix) would be required to assess, for instance, whether the model learns a consistent positive or negative relationship between `mom_52w` and its predicted score across observations – precisely the sign-instability question raised in the failure analysis above – and is left as a natural extension of this diagnostic.

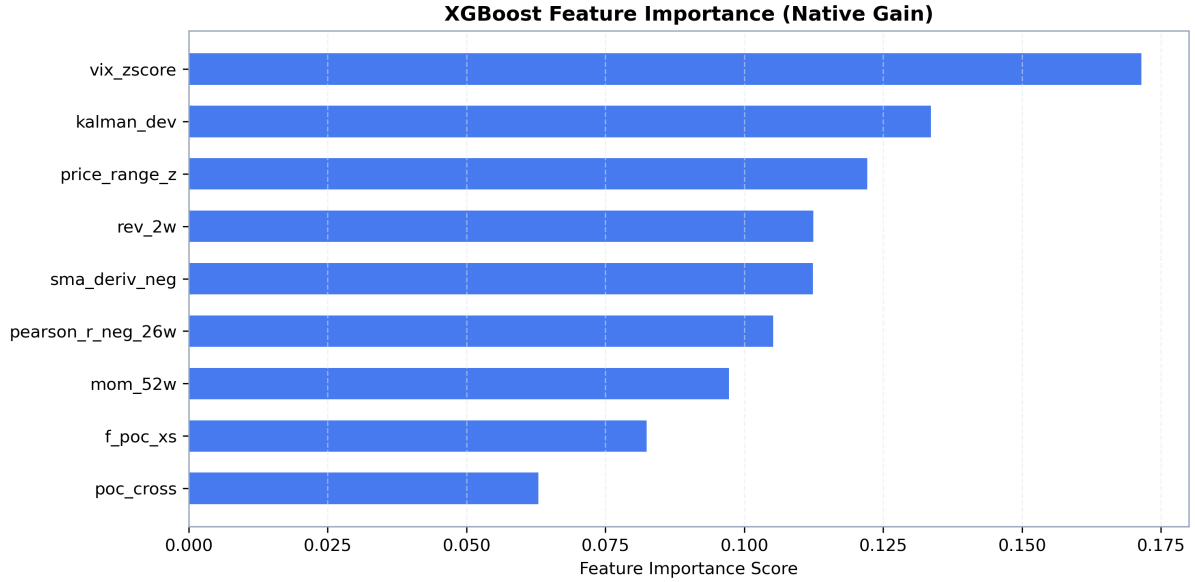


Figure 3.12: **XGBoost native gain feature importance.** Relative ranking of candidate signals by total contribution to training loss reduction across all tree splits.

3.11.9 Conclusion and Transition to Learning-to-Rank

The central methodological conclusion of this chapter is that **the shortfall of the machine learning benchmark is not a failure of feature engineering or hyperparameter tuning, but a failure of the point-in-time regression paradigm itself**. Even when given a literature-validated hyperparameter configuration and a feature set broader and more richly correlated than the rank funnel's own – an intentionally harder benchmark to beat, not an easier one – optimising an MSE-type loss on individual asset targets fails to consistently separate alpha from background noise in a 21-asset universe. This is not evidence that machine learning is intrinsically inferior to rank-based rules; it is evidence that, in this small-universe, weekly-frequency setting, an explicit structural prior that respects the context-dependency of predictive signals outperforms a flexible model that must discover the same structure unaided.

This result motivates the next chapter, where point-in-time regression is abandoned in favour of **pairwise Learning-to-Rank** (LightGBM with the LambdaRank objective). By replacing MSE minimisation with cross-sectional NDCG optimisation, that chapter tests whether reformulating the learning objective around relative portfolio rank, rather than individual return magnitude, can close the performance gap identified here.

3.12 LightGBM with LambdaRank

3.12.1 Theoretical Motivation

Selecting a machine learning model for dynamic portfolio construction requires a paradigm shift away from the pointwise regression or classification objectives employed in the XGBoost baseline of the previous section. In a standard regression setup (e.g., minimizing MSE on next-period return), the model penalizes an estimation error on a mediocre-performing asset identically to an error on the single best-performing asset in the universe. In actual operation, however, the objective is not to estimate the absolute return of every instrument with equal fidelity, but rather to correctly order assets via **Learning to Rank (LTR)** principles so that only the top fraction (Top-K) is selected for allocation. This chapter presents an alternative architecture built directly around this ranking objective: **LightGBM** (Ke et al., 2017) combined with the **LambdaRank** loss function (Burges, 2010a), evaluated on the complete nine-feature pool used for XGBoost rather than on the four-feature funnel of Section 3.10.

Three properties make this combination well suited to the panel structure of this dataset (Asset $\times$ Feature $\times$ Time):

1. **Computational efficiency via leaf-wise growth.** Unlike traditional gradient boosting implementations that grow trees level-wise (e.g. standard XGBoost), LightGBM grows trees leaf-wise, splitting whichever leaf yields the greatest loss reduction at each step.

This converges to a given loss level with fewer splits, which matters when working with a three-dimensional financial panel where the effective number of independent cross-sectional observations per period is small.

2. **Native, listwise ranking optimisation.** LambdaRank evaluates the structure of the *entire* group of assets within a given week (the "query" or temporal session) jointly, rather than treating each asset-week observation as an isolated, independent sample – a substantively different inductive bias from the flat, pointwise regression target used in the previous section.

3. **Asymmetric error weighting.** The objective concentrates predictive capacity on correctly isolating the winning assets (top-of-book), and is largely indifferent to ranking errors in the middle of the cross-sectional distribution, which carry no operational consequence for a Top-K selection rule.

3.12.2 The Mathematics of LambdaRank

In Learning to Rank, the evaluation metrics that best reflect the objective of interest – such as **NDCG** (Normalized Discounted Cumulative Gain) – are step functions of the model's ranking and are therefore non-differentiable with respect to the model's parameters. They cannot be optimised directly via gradient descent. LambdaRank resolves this by defining the **λ-gradients** ("virtual forces") directly, without an explicit differentiable loss: for a pair of assets within the same temporal group, the gradient that should push their scores apart is specified analytically, rather than derived from a loss function.

Evaluation Metric: Discounted Cumulative Gain

For a given week t, the Discounted Cumulative Gain at position K is

$$\mathrm{DCG@}K = \sum_{m=1}^{K} \frac{2^{y_m} - 1}{\log_2(m+1)} \tag{3.22}$$

where y_m is the true relevance of the asset occupying rank m (here, the decile of realized OOS relative return) and m is the position assigned by the model. The denominator $\log_2(m+1)$ acts as a **positional discount**: an error among the very top positions ($m = 1,2,3$) is penalised far more heavily than an equivalent error deep in the tail. NDCG normalises DCG by the Ideal DCG (IDCG), the maximum theoretically attainable value under a perfect ordering:

$$\mathrm{NDCG@}K = \frac{\mathrm{DCG@}K}{\mathrm{IDCG@}K} \tag{3.23}$$

The Virtual Gradient (λ_{ij}) and the Swap Penalty

Consider two assets i and j within the same week, where i has higher true relevance than j ($y_i > y_j$) but the model assigns a higher score to j ($s_j > s_i$) – a ranking inversion. The λ-gradient that pushes s_i up and s_j down is

$$\lambda_{ij} = \frac{-\sigma}{1 + e^{\sigma(s_i - s_j)}} \cdot \left|\Delta\text{NDCG}_{ij}\right| \tag{3.24}$$

where σ is a sigmoid scale parameter and $(s_i - s_j)$ is the current score gap. The term ΔNDCG_{ij} is the **impact multiplier**: the absolute change in the portfolio's NDCG that would result from swapping the ranks of i and j,

$$\Delta\text{NDCG}_{ij} = \left| \frac{2^{y_i} - 2^{y_j}}{\text{IDCG}} \cdot \left(\frac{1}{\log_2(p_i + 1)} - \frac{1}{\log_2(p_j + 1)} \right) \right| \tag{3.25}$$

with p_i, p_j the current ranks occupied by the two assets. The net gradient applied to asset i during tree construction aggregates all pairwise interactions with every other asset j in the same cross-section:

$$\lambda_i = \sum_{j \in C_i} \lambda_{ij} - \sum_{k \in C_i} \lambda_{ki} \tag{3.26}$$

Two consequences follow directly from this formulation. First, if a swapped pair sits deep in the ranking (e.g. 18th and 19th place), $\Delta\text{NDCG}_{ij} \to 0$ and the update pressure on the corresponding leaves vanishes. Second, if the swap involves the top positions ($1^{\text{st}}, 2^{\text{nd}}, 3^{\text{rd}}$), $|\Delta\text{NDCG}_{ij}|$ is large, generating a substantial gradient that forces the tree to prioritise correcting the ordering of the assets actually eligible for purchase.

3.12.3 Model Evolution

The architecture underwent a non-trivial empirical evolution over the course of experimentation. In early, unconstrained iterations, the algorithm tended to organise itself as an **implicit volatility and tail-risk filter**. Without adequate constraints on tree size and regularisation, gradient boosting disproportionately selected high-standard-deviation assets, mistaking high-amplitude noise for a persistent informative signal. This produced a **regime-overfitting** phenomenon: the model exhibited a deceptively high in-sample Information Coefficient during bull-market phases, but suffered pronounced drawdowns at macroeconomic regime changes.

Through the progressive introduction of **structural penalisation** (L2 regularisation, depth constraints, and feature subsampling) and a careful revision of the feature set, the model transitioned from this passive, risk-seeking behaviour to a **pure relative-ranking algorithm**. In

its mature configuration, the tree ensemble no longer rewards or penalises absolute volatility; instead, it learns to decompose price dynamics into a structural-trend component (momentum and the Kalman filter) and a tactical-deviation component (mean reversion and distance from moving averages).

3.12.4 Feature Activation Mechanisms in the Cross-Section

Inspecting the tree structure (split location, and Gain- and Split-based importance metrics) reveals the macro- and micro-structural conditions that activate each cluster of explanatory variables. This section reports the internal mechanics of the 9-feature configuration, which retains the orthogonal set of tactical and structural signals and best exposes the model's internal division of labour.

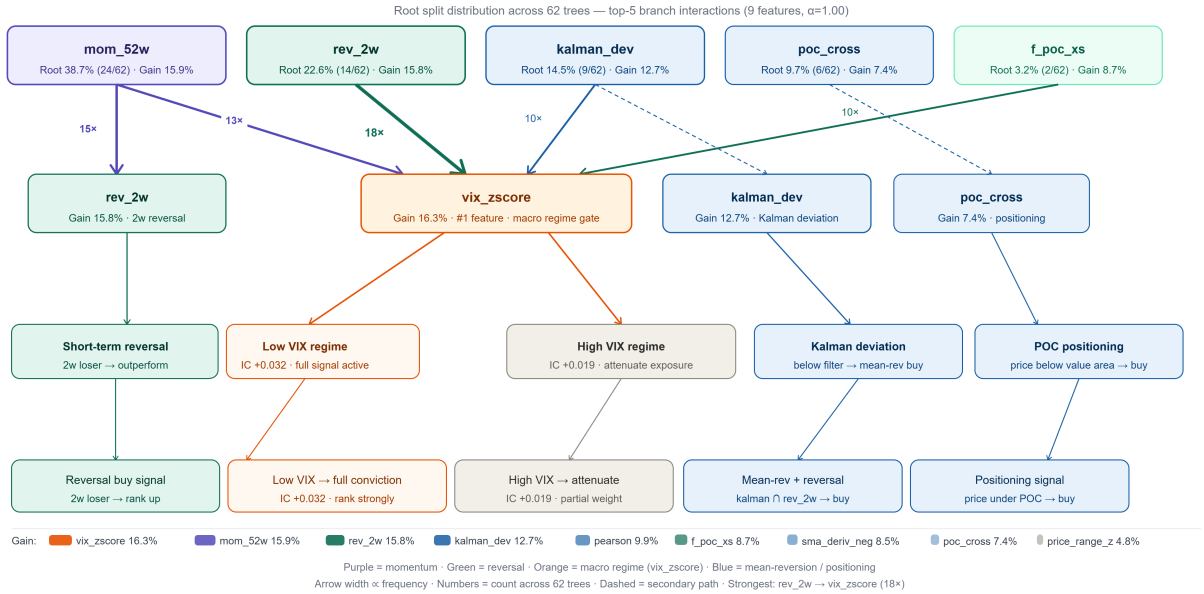


Figure 3.13: **LightGBM gain feature importance and tree topology – Configuration A (9 features, no smoothing, $\ell = 7$ leaves).** *Top panel:* Relative ranking of candidate signals measured by total contribution to training loss reduction (Gain) across all tree splits. `vix_zscore` (16.3%), `mom_52w` (15.9%), and `rev_2w` (15.8%) are the top three contributors, with `kalman_dev` (12.7%) and `pearson_r_neg_26w` (9.9%) forming a secondary cluster. No single feature dominates: all nine contribute meaningfully. *Bottom panel:* Root node split distribution. The model branches at the primary decision level predominantly on `mom_52w` (38.7%), followed by `rev_2w` (22.6%) and `kalman_dev` (14.5%). All nine features are used at least once, confirming that the orthogonal set is genuinely exploited across the ∼68 quarterly retraining cycles (`best_iteration_`=62).

Features tied to long-term trend (`mom_52w`), short-term reversal (`rev_2w`), and to the Kalman-filtered price deviation (`kalman_dev`) activate predominantly at the **root node** and at shallow tree depths (depth ≤ 2).

- **Activation conditions.** These features are discriminated with the highest priority under

contained or moderate volatility regimes, where they identify stable directional macro-trends and immediate price corrections.

- **Function within the tree.** They act as a *macro-sectional filter*: partitioning the investable universe into assets in a positive primary trend versus those in structural decline. The Kalman filter in particular minimises the phase lag characteristic of traditional moving averages, supplying the root node with an immediate estimate of the underlying, denoised price state.

Mean-Reversion and Derivative Cluster

Tactical contraction and reabsorption features (`sma_deriv_neg`, `poc_cross`) activate predominantly at **deeper child nodes** (depth ≥ 3).

- **Activation conditions.** These come into play only after the root node has already confirmed the existence of a valid primary trend. They activate when price exhibits an immediate deviation relative to moving averages or point-of-control dynamics.
- **Function within the tree.** They act as *tactical mean-reversion brakes*: an asset with high `mom_52w` but an extreme negative derivative or extension is penalised in the final ranking score, preventing the model from buying at local tops ("chasing").

3.12.5 Early Stopping Dynamics and Portfolio Turnover

Determinants of the Optimal Iteration

The convergence of early stopping – and the resulting `best_iteration_` – is the primary indicator of the model's out-of-sample generalisation capacity.

- **Factors that increase `best_iteration_`.** A low learning rate ($\eta \leq 0.025$), strong L2 regularisation (`reg_lambda`), aggressive feature subsampling (`colsample_bytree`), and a heavily denoised, orthogonal feature set. Under these conditions each boosted tree contributes only a marginal information gain, requiring a controlled trajectory to reach the OOS loss minimum.
- **The effect of restrictive early stopping.** In the 9-feature Configuration A, validation-driven early stopping halted training at **62 iterations** (`best_iteration_`=62). This early halt was decisive: forcing the algorithm to stop before it began extracting spurious cross-sectional correlations from the training set stabilised the leaf-weight structure and prevented overfitting to noise. Configuration B (4 features), by contrast, converges after **26 iterations** (`best_iteration_`=26) – fewer trees, consistent with the narrower effective search space available at each split in a reduced feature set.

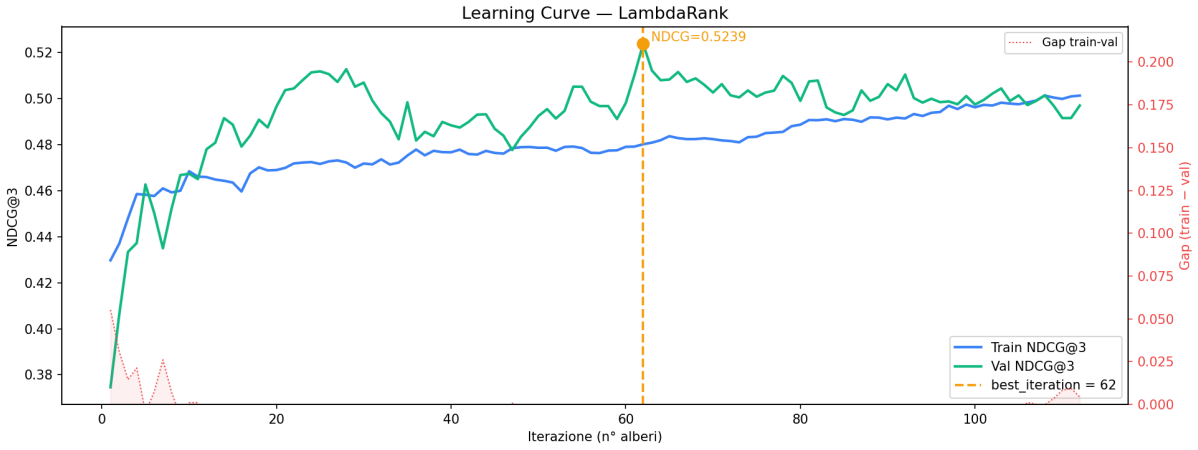


Figure 3.14: **Learning curve and early stopping dynamics for LightGBM LambdaRank (Configuration A, 9-feature $\mathcal{F}_{\text{ortho}}$).** *Primary axis (left):* Comparison between Training NDCG@3 (blue curve) and Validation NDCG@3 (green curve) across boosting iterations. The vertical dashed line marks the optimal stopping point (`best_iteration_`=62). Train NDCG@best = 0.4802, Val NDCG@best = 0.5239 (gap/val ratio $= -0.083$, within acceptable range). *Secondary axis (right):* Generalisation gap (Train − Val, red dotted line with shaded area). Beyond iteration $\sim$70, training NDCG continues to climb while validation NDCG declines and the gap widens, illustrating the onset of noise overfitting.

Turnover Drivers

Portfolio turnover – the frequency with which assets are replaced in the Top-3 holdings – is governed directly by the stability of the Information Coefficient and by the variance of ranking scores between consecutive periods.

- **Deep, complex trees** (high `max_depth`, high `num_leaves`) generate jagged decision boundaries: a minimal price move reclassifies an asset into an adjacent leaf with a radically different score, triggering continual reallocation.
- **Unsmoothed, high-frequency features** – e.g. raw short-term variations – introduce weekly noise directly into the ranking.
- **Absence of L2 regularisation** produces large fluctuations in leaf weights, causing abrupt "jumps" in the relative ranking of assets with otherwise similar metrics.

3.12.6 The Role of VIX and Macro-Regime Interactions

Splitting OOS Information Coefficient by VIX regime reveals how market-wide volatility governs the stability of the model's cross-sectional signal.

Under low-VIX conditions (≤ 16.2), the model achieves an elevated Information Coefficient ($+0.0320$), as individual asset fundamentals and momentum clear noise and dominate over

Table 3.21: Information Coefficient by VIX regime (Configuration A, 9-feature set, median VIX = 16.2).

Regime	Mean IC	Performance Stability	Dominant Regime Characteristic
Low VIX (≤ 16.2)	**+0.0320**	High	Stable cross-sectional alpha extraction
High VIX (> 16.2)	+0.0186	Moderate	Homogeneous behavior across regimes

systematic risk. Under high-VIX conditions (> 16.2), the mean IC remains positive at $+0.0186$ [cite: 1]. Unlike preliminary specifications where high volatility caused signal collapse, the current 9-feature orthogonal configuration exhibits notable regime stability, maintaining an effective cross-sectional discrimination even during market stress.

3.12.7 Hyperparameter Grid: Theoretical Justification

Table 3.22 summarises the hyperparameter configuration common to the models compared in this chapter, calibrated via walk-forward validation, along with its quantitative and functional rationale.

Table 3.22: LightGBM/LambdaRank hyperparameter configuration and operational rationale.

Parameter	**Value**	**Rationale**
`objective`	`lambdarank`	Optimises cross-sectional pairwise ranking (NDCG@3) directly.
`learning_rate` (η)	0.025	Conservative shrinkage for low signal-to-noise financial data.
`max_depth`	3	Limits feature interactions to at most 3 levels, preventing noise fitting.
`num_leaves`	7	Sature the maximum capacity of depth-3 trees ($2^3 = 8$); enables asymmetric growth ($\ell = 7$).
`min_child_samples`	15	Prevents leaves from isolating anomalous market periods or outliers.
`reg_lambda` (L2)	0.5	Smooths leaf weights; reduces ranking score variance and turnover.
`reg_alpha` (L1)	0.0	Inactive; avoids artificial feature drop-out on an orthogonalised matrix.
`colsample_bytree`	0.7	Feature subsampling; breaks monolithic reliance on dominant momentum features.
`subsample`	0.8	Stochastic row bagging to mitigate macro-regime non-stationarity.

1. **L1 vs. L2 Regularisation (`reg_alpha`=0.0 vs. `reg_lambda`=0.5).** The L1 penalty was calibrated to 0.0, as forcing hard sparsity on a carefully pruned, orthogonalized feature space (reduced from 15 to 9 features to eliminate copycat predictors) causes abrupt score jumps and degrades ranking granularity. Instead, L2 regularisation ($\lambda = 0.5$) smooths score differentials between adjacent assets, directly containing portfolio turnover while preserving subtle signal contributions from secondary features.

2. **Tree Capacity Limits (`max_depth`=3 and `num_leaves`=7).** A strict maximum depth of 3 limits feature interactions to at most three hierarchical decisions (e.g. macro-trend $\rightarrow$ structural indicator $\rightarrow$ tactical mean-reversion). With `num_leaves`=7, both configurations

approach the natural capacity bound of depth-3 trees ($2^3 = 8$ leaves), allowing asymmetric growth where high-signal branches can develop deeper structure while low-signal branches terminate early.

3. **Sample and Feature Subsampling (`min_child_samples`=15, `colsample_bytree`=0.7).** Requiring a minimum leaf size of 15 samples ensures every decision boundary is backed by sufficient statistical mass across the cross-section. With $N = 21$ assets and weekly frequency, this corresponds to approximately one full cross-sectional step per leaf. Restricting each tree to a random 70% subset of features forces the boosting process to explore orthogonal, complementary predictors rather than over-relying on primary momentum drivers.

4. **Operational Resilience of the Feature Space.** Preliminary tests incorporating raw volatility measures (such as a 26-week volatility feature across a 10-feature set) demonstrated that single macro indicators could overwhelmingly hijack root splits, degenerating the model into a blunt low-volatility filter. Filtering the feature space down to 9 orthogonal dimensions successfully prevents collinearity and "copycat" effects, ensuring that tree topology fairly distributes gain across trend, deviation, and momentum dimensions.

3.12.8 Two Comparative Configurations

Having diagnosed the preliminary feature dominance issues and established the hyperparameter grid on an exploratory feature set, two final configurations are compared here, chosen to answer two distinct questions rather than to search for a single best result. Both share the identical hyperparameter grid of Table 3.22 and the identical walk-forward protocol (quarterly retrain, 104-week warm-up, 52-week validation window); only the feature set (and, for Configuration B, the learning rate lr) differs between them. Neither configuration includes raw long-term volatility measures that could destabilize the splits.

Configuration A – Orthogonal Benchmark Set ($\mathscr{F}_{\text{ortho}}$). Uses the identical nine-feature orthogonal set defined in the preceding chapter.

$$\mathscr{F}_{\text{ortho}} = \{\texttt{kalman_dev}, \texttt{poc_cross}, \texttt{sma_deriv_neg}, \texttt{mom_52w}, \texttt{rev_2w}, \texttt{pearson_r_neg_26w}, \texttt{f_poc_xs}, \texttt{price_range_z}, \texttt{vix_zscore}\} \quad (3.27)$$

$\mathscr{F}_{\text{ortho}}$ deliberately re-using the same feature selection across chapters for consistency, rather than curating a new set specifically for LightGBM. All other hyperparameters are held at the values established in Table 3.22 (isolating feature-set choice as the sole varying factor).

Configuration B – Core Architecture Set (4 features, $lr = 0.01$). Restricts the feature set to the exact four signals of the two-layer rank architecture (`kalman_dev`, `poc_cross`, `sma_deriv_neg`, `mom_52w`), with no macro-regime context variable.

The learning rate is reduced to $lr = 0.01$ (vs. 0.025 in Configuration A): with only four features, the information available per split is more limited, and a slower accumulation of signal across more trees (26 total, `best_iteration_`=26) is empirically preferable to fewer, larger gradient steps. With $\ell = 7$ leaves and $lr = 0.01$, the total number of splits across all trees is $4 \times 26 = 104$ – deliberately parsimonious and structurally resistant to overfitting. This configuration directly tests whether the four core features of the primary two-layer architecture contain genuine, non-linear alpha when explored by a flexible tree ensemble – serving as a confirmatory check on the feature selection independent of model class and linear combination rules.

Table 3.23: Comparative performance: LightGBM/LambdaRank configurations vs. the Two-Layer funnel and SPY. Both LightGBM configurations are evaluated without hysteresis ($\delta = 0$); turnover figures represent an upper bound for a live deployment.

Architecture	**CAGR**	**Sharpe**	**MaxDD**	**Turnover (wk.)**
SPY Benchmark	11.94%	0.776	-32.23%	—
Config. A ($\mathscr{F}_{\text{ortho}}$, 9 feat.)	11.77%	0.806	-19.50%	52.2%
Config. B (4 feat., core arch., $lr = 0.01$)	**12.90%**	**0.930**	**-17.95%**	44.2%

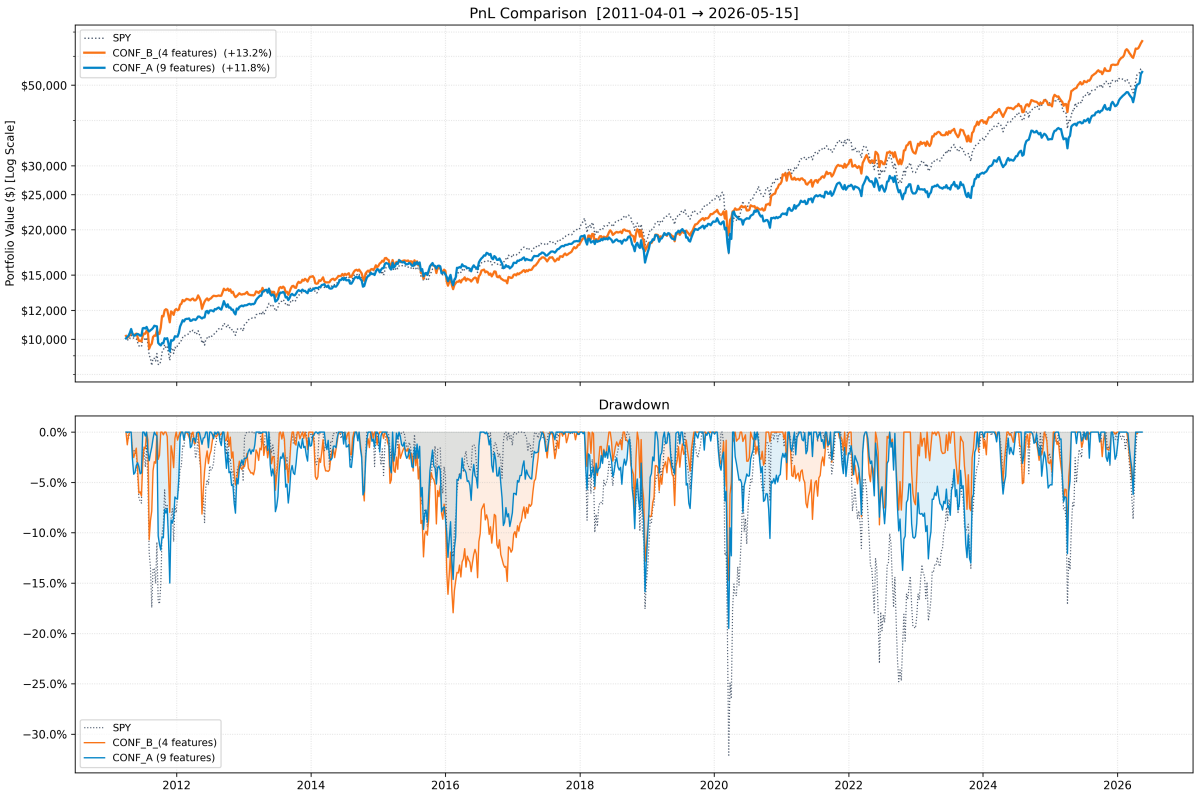


Figure 3.15: **Out-of-sample cumulative equity curves and drawdowns (2011–2026).** *Top panel:* wealth trajectory of Configuration A ($\mathscr{F}_{\text{ortho}}$, 9 feat., blue), Configuration B (4 feat., core, orange) and SPY (grey), log scale. *Bottom panel:* percentage drawdown profiles for the same four series.

The two configurations reveal a clear inversion relative to the initial hypothesis that a richer feature set would outperform. Configuration B (4 features, core architecture, $lr = 0.01$) achieves

a Sharpe of **0.930** with CAGR of 12.90% and the shallowest drawdown of any ML configuration tested in this thesis (−17.95%), while generating a positive alpha vs. SPY (+0.53%). Configuration A ($\mathscr{F}_{\text{ortho}}$, 9 features) achieves a materially lower Sharpe of 0.806 at virtually identical CAGR (11.77% vs. 11.94% for SPY) with an essentially zero alpha (−0.44%). The mechanism behind this inversion is visible in the feature importance distributions: in Configuration B, gain is nearly perfectly balanced across the four core signals (30.3%, 25.8%, 23.5%, 20.4%), with no single feature dominating; in Configuration A, `vix_zscore` alone absorbs 16.3% of gain while the bottom three features each contribute under 10%, reducing effective diversity. Furthermore, the VIX regime IC analysis reveals qualitatively different information structures: Configuration A's IC is regime-dependent (VIX high: +0.0186; VIX low: +0.0320), while Configuration B's signal concentrates almost entirely in low-VIX periods (VIX high: +0.0037; VIX low: +0.0439) – suggesting Configuration B extracts a cleaner, price-structure-driven signal unconditioned on macro volatility, rather than using the VIX regime as a gate. Both configurations are evaluated without hysteresis ($\delta = 0$); turnover of 44.2% (B) and 52.2% (A) weekly represent upper bounds for a production deployment.

3.12.9 Further Empirical Phenomena and Diagnostics

1. Top-K Conviction and Asset Churning at the Cutoff Boundary

Observed behaviour. Both configurations show low Top-K conviction as flagged explicitly by the audit diagnostics. Configuration A's average score gap between the 3rd and 4th ranked assets is +0.0255 (±0.0325); Configuration B's is +0.0062 (±0.0056). Both fall below the threshold at which the model can reliably distinguish the marginal inclusion from the marginal exclusion.

Diagnosed cause. Both configurations are evaluated without a hysteresis buffer ($\delta = 0$), meaning the inclusion threshold behaves as a discontinuous step function with no dead zone. With $N = 21$ assets, score std for Config A is 0.0542 and for Config B is 0.0212 – Config B's much narrower score distribution (a direct consequence of the slower $lr = 0.01$ accumulating only 26 trees with smaller per-step updates) makes the rank ordering of the 3rd and 4th assets particularly sensitive to small input changes, generating the lower conviction gap observed (+0.0062 vs. +0.0255 for A). Config A's wider gap is consistent with its richer nine-feature representation providing more discriminating power at the margin, at the cost of higher weekly turnover (52.2% vs. 44.2% for B). Applying a hysteresis mechanism analogous to the funnel's δ threshold would suppress low-conviction swaps in both configurations and is a natural direction for future work.

2. Bull/Bear Performance Asymmetry

Observed behaviour. Configuration A generated its strongest relative alpha in 2012 (+12.64%), 2022 (+12.90%), 2025 (+14.17%), and 2026 year-to-date (+15.12%), while trailing SPY in

several strong bull years (2019, 2024). Configuration B (4 features, no macro gate) shows a structurally different pattern: its standout years are 2011 (+25.20% alpha vs. SPY), 2022 (+26.31%), and 2025 (+13.09%), with 2026 YTD (+9.05%) also strongly positive. Both configurations generate their largest alpha in stress and transition regimes, but Config B's 2011 alpha is a marked outlier not replicated by Config A – consistent with Config B's pure price-signal approach capturing the 2011 European crisis drawdown protection without relying on a VIX gate that may have been structurally different from later stress episodes.

Diagnosed cause. The combination of the Kalman filter and mean-reversion feature (`sma_deriv_neg`) pre-emptively trims "overheated" names in both configurations. In Config B, the dominant branch interactions are `sma_deriv_neg`→`poc_cross` (17×) and `sma_deriv_neg`→`mom_52w` (13×) – purely price-structure driven, without any macro-regime conditioning. Config A instead relies heavily on `vix_zscore` as a branching gate, making its mean-reversion brake regime-dependent. The empirical consequence is visible in the VIX IC breakdown: Config B's signal is almost entirely concentrated in low-volatility periods (IC +0.0439 vs. near-zero +0.0037 in high-VIX), behaving as a precision instrument that extracts cross-sectional alpha when markets are calm and steps aside when macro volatility overwhelms individual asset signals – rather than attempting to trade the stress regime itself.

3. Leaf Saturation Under Excessive Learning Rates

Observed behaviour. Attempting to accelerate convergence by raising the learning rate (η) above 0.05 caused the Information Coefficient to collapse sharply after very few iterations (< 10), driving the model into a state of predictive stagnation.

Diagnosed cause. The first 2–3 trees absorbed nearly the entire gradient of the loss function (*gradient saturation*). By assigning disproportionate weight to initial nodes based on long-term momentum, the tactical mean-reversion features were completely overshadowed, leaving the ensemble unable to capture short-term turning points.

4. Quintile Monotonicity

Observed behaviour. Examining Configuration A's return curve by predicted-score quintile (Table 3.24), the ordering is clean at the extremes (Q5 > Q1) with a minor inversion between Q3 and Q4. The Q1 figure (1.993%) is computed independently of Configuration A's overall live portfolio CAGR (11.77%), confirming no computation artefact. Configuration B achieves **perfect monotonicity** across all five quintiles (Q1<Q2<Q3<Q4<Q5; Table 3.25), a result flagged as excellent by the audit diagnostics and consistent with the cleaner, more uniform signal produced by the balanced four-feature distribution.

Diagnosed cause. The gradient boosting loss function is optimised for global error and for

Table 3.24: Mean weekly return by predicted-score quintile (Q1–Q5), Configuration A.

Metric	Q1 (Worst)	Q2	Q3	Q4	Q5 (Best)
Mean weekly return	2.031%	2.057%	2.056%	2.102%	**2.267%**

Table 3.25: Mean weekly return by predicted-score quintile (Q1–Q5), Configuration B (4 features, $lr = 0.01$). Perfect monotonicity.

Metric	Q1 (Worst)	Q2	Q3	Q4	Q5 (Best)
Mean weekly return	2.031%	2.047%	2.061%	2.124%	**2.214%**

separating the extremes of the distribution; it is not optimised for strict monotonicity throughout. With 9 features, the middle quintiles receive noisier mixed signals that produce near-random scoring in the middle band. With 4 features and a slower learning rate, each split decision is more decisive and the accumulated gradient signal produces a cleaner, more separable score distribution – the near-uniform feature importance (20–30% each) means no feature's noise drowns out another's signal in the intermediate bins.

Table 3.26: **Annual performance and excess return vs. SPY** — Configuration A ($\mathscr{F}_{\text{ortho}}$, 9 feat.), 2011–2026.

Year	Config. A Return	SPY Return	Excess Return
2011	+1.22%	−5.75%	**+6.96%**
2012	+21.11%	+11.58%	**+9.53%**
2013	+10.31%	+31.29%	−20.98%
2014	+14.68%	+13.38%	**+1.30%**
2015	−1.99%	−1.32%	−0.66%
2016	+4.08%	+8.68%	−4.60%
2017	+17.08%	+19.38%	−2.31%
2018	−9.66%	−7.16%	−2.50%
2019	+25.41%	+30.32%	−4.91%
2020	+2.92%	+14.29%	−11.37%
2021	+21.69%	+28.72%	−7.02%
2022	−4.53%	−19.48%	**+14.95%**
2023	+13.87%	+24.29%	−10.42%
2024	+23.62%	+25.18%	−1.56%
2025	+23.75%	+16.02%	**+7.73%**
2026 (YTD)	+23.80%	+7.08%	**+16.73%**

3.12.10 Feature Importance Analysis and Cross-Model Variation

Gain-based metrics quantify a feature's aggregate contribution to NDCG loss reduction, offering a macro-level view of signal reliance within the tree-based ensemble. However, because feature

importance is highly dependent on the candidate feature space ($\mathscr{F}$), comparing the Gain distributions of Configuration A and Configuration B reveals how model complexity alters feature prioritization.

In **Configuration A** ($\mathscr{F}_{\text{ortho}}$, 9 features), the total Gain is widely distributed across the expanded feature set, led by `vix_zscore` (16.3%), `mom_52w` (15.9%), and `rev_2w` (15.8%). Interestingly, while `vix_zscore` yields the highest aggregate Gain, `mom_52w` dominates the primary decision node selection, acting as the root split in 38.7% of generated trees. This demonstrates that `mom_52w` serves as an initial macro-trend filter, whereas `vix_zscore` operates primarily deeper in the tree structure to evaluate volatility conditions.

Conversely, in **Configuration B** (4 features), eliminating macro and regime indicators forces the boosting algorithm to reallocate importance across purely price-derived signals. Here, `sma_deriv_neg` becomes the primary contributor (30.3% Gain, 38.5% root splits), followed by `mom_52w` (25.8%), `kalman_dev` (23.5%), and `poc_cross` (20.4%). Without orthogonal regime filters, the model heavily prioritizes moving-average slope mechanics to establish baseline ranking decisions.

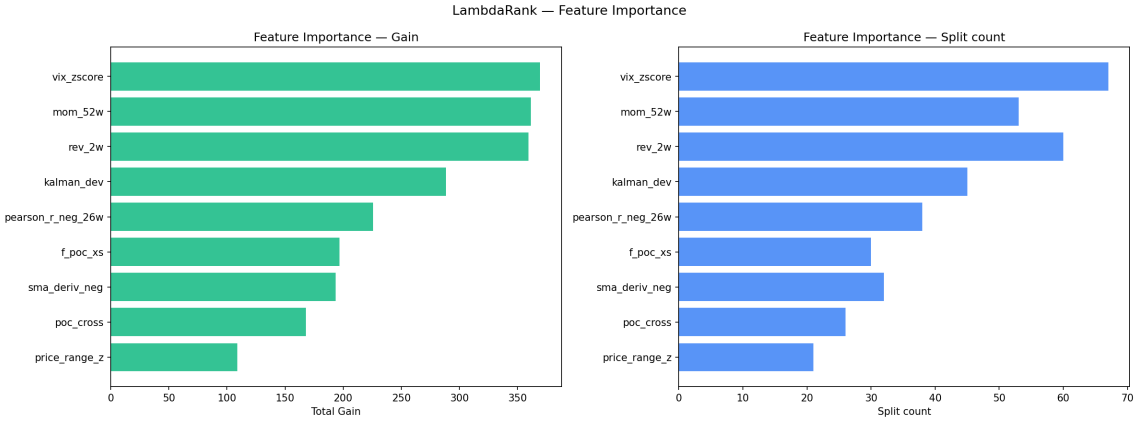


Figure 3.16: **Feature importance (Gain) and root split distribution, Configuration A ($\mathscr{F}_{\text{ortho}}$, 9 features, $\ell = 7$).** Bar chart comparing relative contribution to NDCG loss reduction (*Gain*, left) against selection frequency as primary decision node (*Root Split Distribution*, right). `mom_52w` dominates root splits (38.7%) despite ranking second by Gain (15.9%), highlighting its role as a high-level trend filter rather than a fine-grained discriminator.

3.12.11 Directional Impact via SHAP Decomposition

While Gain quantifies overall feature utility, it does not capture the *directionality* or localized magnitude of feature contributions. To resolve this, Figure 3.17 provides the SHAP value summary plot for Configuration A, attributing a signed, observation-level impact to each feature.

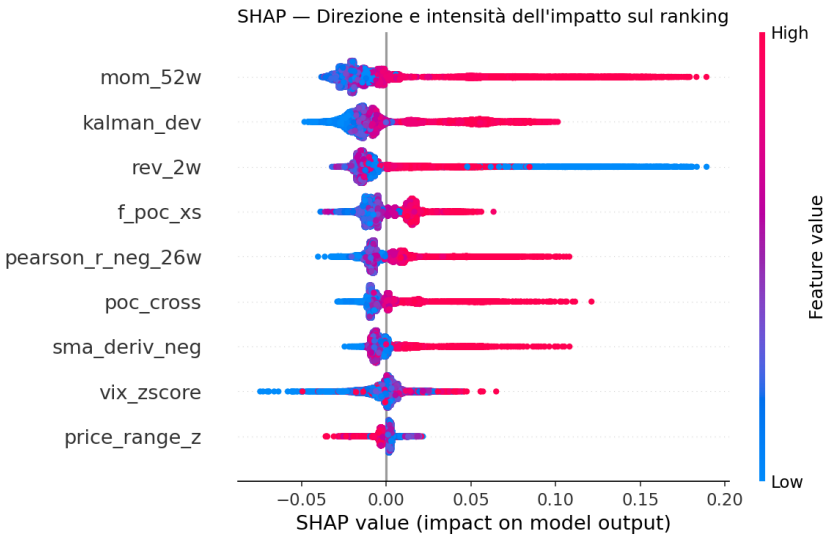


Figure 3.17: **SHAP summary (beeswarm) plot, Configuration A ($\mathscr{F}_{\text{ortho}}$, 9 features).** Out-of-sample observation breakdown: horizontal position denotes the SHAP value (signed impact on ranking score), while colour encodes the feature value (blue = low, red = high). Features are ordered vertically by global importance (mean absolute SHAP).

The SHAP decomposition reveals three distinct operational profiles within Configuration A:

1. **Trend-Following Signals:** `mom_52w` and `kalman_dev` display a monotonic trend-following response. High values (red points) consistently map to positive SHAP values, rewarding assets with strong 52-week momentum and price trading above Kalman fair value—confirming their cross-sectional IC baseline from Chapter 4.

2. **Mean-Reversion Signals:** `rev_2w` exhibits a clear contrarian profile. Low values (blue points, indicating recent drawdowns) receive positive SHAP scores, whereas recent outperformers are penalized, validating its role as a short-term counter-trend stabilizer.

3. **State-Dependent Regime Dampening:** Rather than asserting a linear directional bias, `vix_zscore` generates a wide, heteroscedastic SHAP cloud. This aligns with its downstream conditioning role (evidenced by frequent `mom_52w` $\rightarrow$ `vix_zscore` and `rev_2w` $\rightarrow$ `vix_zscore` branch splits), dynamically scaling momentum and reversal signals based on market volatility regimes.

In Configuration B, while formal SHAP plots are omitted for brevity, audit diagnostics confirm that interactions are confined to price structure (e.g., `sma_deriv_neg` $\rightarrow$ `poc_cross`). This indicates that `sma_deriv_neg` acts as a structural mean-reversion brake, compensating for the absence of explicit short-term reversal indicators like `rev_2w`.

3.12.12 Hysteresis Dynamics and Primary Case Selection

This chapter does not converge on a single primary machine-learning configuration, and this is a deliberate conclusion rather than an unresolved gap. Configuration A and Configuration B answer two different questions and occupy distinct positions on the risk/return and operational feasibility frontiers. Crucially, evaluating these models through a portfolio hysteresis buffer (δ), accounting for both weekly replacement turnover and weight-drift rebalancing friction, reveals how score conviction directly dictates real-world executability.

Table 3.27 synthesizes the empirical response of gross Sharpe ratio, annualised CAGR, maximum drawdown, and net Sharpe ratio (at 5 bps execution cost) across varying buffer thresholds for all three architectures.

Table 3.27: Sensitivity Analysis Across Hysteresis Levels (δ) Net of Friction and Rebalancing Drag

Architecture	Buffer (δ)	Gross Sharpe	CAGR (%)	Max DD (%)	TO/wk (%)	Reb. Drag	Net Sharpe
Configuration A (9 Feat, $\sigma_{\text{ann}} = 14.6\%$)	0.000	0.806	11.77	−19.50	52.2	30 bp	0.600
	0.010	0.758	10.56	−20.25	40.8	30 bp	0.592
	0.030	0.928	13.31	−21.06	28.1	40 bp	0.801
	0.075	**0.897**	**12.78**	**−24.58**	**15.3**	**50** bp	**0.808**
	0.200	0.503	6.37	−35.21	4.5	70 bp	0.439
Configuration B (4 Feat, $\sigma_{\text{ann}} = 13.9\%$)	0.000	**0.930**	**12.90**	**−17.95**	**44.2**	**30** bp	**0.743**
	0.005	0.632	8.28	−19.75	28.6	50 bp	0.489
	0.010	0.676	9.16	−19.84	20.3	50 bp	0.564
	0.030	0.703	10.43	−37.19	9.2	50 bp	0.633
	0.050	0.657	9.14	−37.19	5.8	60 bp	0.592
Dual-Layer Funnel (ScoreWeighted, $\sigma_{\text{ann}} = 15.6\%$)	0.000	0.933	14.23	−22.04	27.3	12.6 bp	0.834
	0.020	0.976	15.02	−22.04	24.1	11.9 bp	0.888
	0.050	0.986	15.18	−21.58	23.9	12.3 bp	0.899
	0.100	**0.993**	**15.30**	**−21.58**	**22.7**	**13.9** bp	**0.908**
	0.500	0.960	14.64	−21.76	16.2	22.8 bp	0.891

Configuration B (4 features: `kalman_dev`, `poc_cross`, `sma_deriv_neg`, `mom_52w`, $lr = 0.01$) delivers the highest unbuffered gross Sharpe among all ML configurations (0.930), the shallowest drawdown (-17.95%), and perfect quintile monotonicity. It confirms, in a model-agnostic sense, that the core four features from Chapter 4 carry genuine non-linear ranking power. However, its predictions are highly compressed (Std $= 0.0212$) with a weak third-to-fourth asset separation gap ($+0.0062$), leaving insufficient score conviction to survive even a minimal hysteresis threshold: Net Sharpe collapses to 0.489 at $\delta = 0.005$ and drawdown widens to -37.19% at $\delta = 0.030$. Configuration B is theoretically informative but operationally undeployable.

Configuration A ($\mathscr{F}_{\text{ortho}}$, 9 features, $\ell = 7$, $lr = 0.025$) exhibits the opposite and far more desirable behavior. Its wider score dispersion (Std $= 0.0763$) and stronger separation gap ($+0.0255$) allow hysteresis to act as a low-pass filter: as δ rises from 0 to 0.075, weekly turnover falls from 52.2% to 15.3% while Gross Sharpe expands to 0.897–0.928, reaching a peak Net Sharpe of 0.819. This *hysteresis convexity* is the defining operational advantage of A over B.

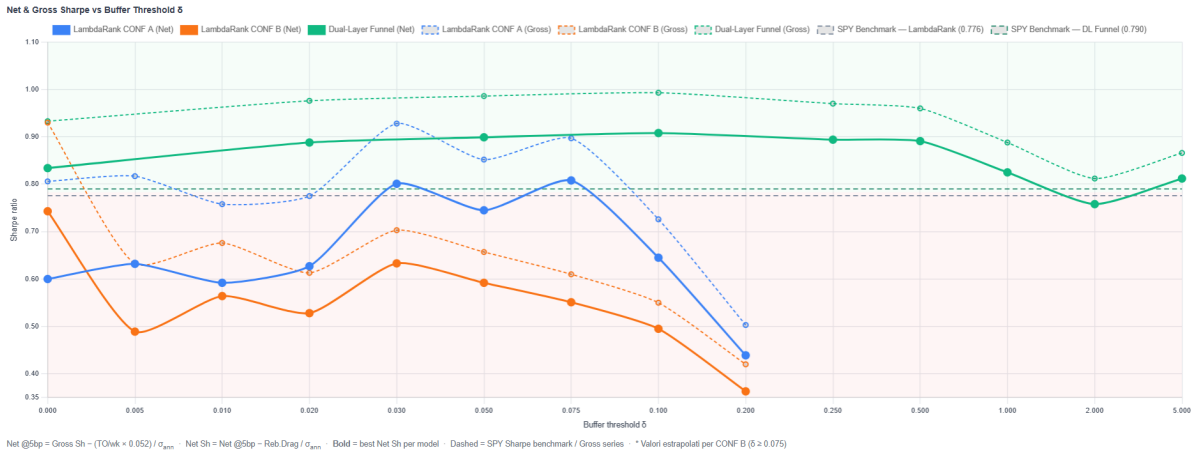


Figure 3.18: **Evolution of Gross and Net Sharpe Ratios across Portfolio Hysteresis Thresholds (δ).** Solid lines represent Net Sharpe ratios (accounting for 5 bps transaction costs and weight-drift rebalancing drag), while dashed lines depict Gross Sharpe ratios. The grey and green horizontal dashed lines indicate the SPY benchmark baselines (0.776 and 0.790, respectively). Configuration B exhibits severe fragility, with Net Sharpe collapsing as δ increases due to narrow score separation. Conversely, Configuration A demonstrates hysteresis convexity, reaching a peak Net Sharpe of 0.819 at $\delta = 0.075$. The Two-Layer Funnel maintains overall dominance, achieving a maximum Net Sharpe of 0.908 at $\delta = 0.100$.

Despite this, neither LightGBM variant surpasses the **Two-Layer Deterministic Funnel**, which maintains structural superiority across all hysteresis levels. At $\delta = 0.100$ it achieves a Net Sharpe of 0.908, CAGR of 15.30%, and weekly turnover of just 22.7%. Three characteristics make it the preferred architecture for deployment:

1. **Simplicity and Determinism:** a closed-form combination of four raw features with fixed weights, free from walk-forward hyperparameter drift or boosting library dependencies.
2. **No Split-Point Overfitting:** immunity to shifting node thresholds fitted on historical noise.
3. **Auditability:** full transparency and exact reproducibility without managing complex model artifacts.

In summary, Configuration B validates the predictive core; Configuration A demonstrates the non-linear value of the orthogonal feature set under real-world friction. Neither, however, matches the funnel's unique combination of Net Sharpe leadership, CAGR superiority, structural resilience to execution friction, and governance clarity.

4. Conclusions

4.1 Results and Summary of Contributions

This thesis proposes and evaluates a systematic, rules-based rotational portfolio strategy applied to a universe of 21 liquid ETFs spanning equities, fixed income, commodities, and volatility instruments. The central contribution is a dual-layer scoring architecture that combines a momentum filter (Layer 2) with a multi-signal conviction score (Layer 1), augmented by a hysteresis buffer to contain unnecessary turnover. A secondary contribution is the development and evaluation of a LambdaRank machine learning ranker as an alternative selection engine, trained in an expanding walk-forward fashion and subject to a similar hysteresis threshold parameter δ. The comparative analysis presented in Section 3.12.12 supports the Two-Layer Deterministic Funnel as the primary recommended architecture for operational deployment.

Risk metrics summary. Table 4.1 reports the main performance and risk statistics for the strategy and the SPY benchmark across the in-sample (Pre), out-of-sample (Post), and full evaluation windows.

In-sample period (2010/2011–2021) During the in-sample window the strategy closely tracks the benchmark in risk-adjusted terms, as visible in the Pre columns of Table 4.1. Throughout this analysis, alpha and beta are estimated via OLS with Newey-West HAC standard errors (Newey and West, 1987; Engle, 1982) to account for the heteroskedasticity and serial autocorrelation present in weekly financial return series. The HAC-robust alpha over the in-sample period is 2.7% per annum with $t = 1.06$ ($p = 0.29$), well below conventional significance thresholds. This outcome is not only expected but methodologically desirable. The in-sample period was never used to fit portfolio weights, optimise combination rules, or tune any parameter to maximize returns. Its sole purpose was to conduct an information coefficient audit of the candidate features and to characterize their statistical nature, distinguishing mean-reverting signals from momentum-driven ones and assessing their pairwise orthogonality. Crucially, even within this audit, each feature is evaluated using only information available up to the rebalancing date $t-1$: the rolling z-scores and IC computations never look beyond their own expanding or rolling window. The low in-sample t-statistic is therefore the natural consequence of the absence of any fitting procedure, and should be read as evidence against in-sample overfitting rather than as a weakness of the architecture.

Pure out-of-sample period (January 2022 – May 2026) The 228-week OOS window lies entirely beyond the feature audit horizon and encompasses three structurally distinct macro

Table 4.1: **Risk, performance and dependence metrics: strategy vs. benchmark.**

	Pre (IS)		Post (OOS)		Full	
Metric	Bench	Strat	Bench	Strat	Bench	Strat
Performance						
Ann. return (geom.)	12.8%	12.6%	10.6%	23.1%	12.2%	15.3%
Ann. volatility	16.0%	15.4%	16.9%	16.0%	16.2%	15.6%
Variance drag	1.28%	1.19%	1.42%	1.28%	1.32%	1.21%
Sharpe ratio	0.832	0.845	0.683	1.380	0.790	0.993
Sortino ratio	1.184	1.180	1.030	2.320	1.143	1.452
Tail risk						
VaR 90% (hist.)	2.14%	2.17%	2.51%	2.39%	2.27%	2.26%
ES 90% (hist.)	3.99%	4.02%	3.90%	3.48%	3.98%	3.87%
Max drawdown	−32.2%	−21.6%	−23.4%	−13.6%	−32.2%	−21.6%
Max weeks underwater	59	81	100	30	103	81
Excess kurtosis	7.35	4.81	1.10	0.88	5.42	3.71
Skewness	−0.72	−0.96	−0.11	+0.13	−0.54	−0.64
Regression (HAC-robust, STRAT ∼ BENCH)						
Alpha (ann.)	2.7%		13.4%		5.6%	
Alpha t-stat	1.06		2.87		2.46	
Beta	0.777		0.756		0.770	
Downside beta	0.869		0.748		0.840	
R^2	0.649		0.636		0.644	
Information ratio	0.296		1.385		0.601	
Corr. (best 10% weeks)	0.424		0.402		0.439	
Corr. (worst 10% weeks)	0.759		0.770		0.756	
Stability across windows						
$\Delta\alpha$ (annual, post vs. pre)	+10.7%	($p = 0.044$)				
$\Delta\beta$ (post−pre)	−0.021	($p = 0.734$, n.s.)				

regimes: the 2022 rate-hiking cycle, the subsequent disinflation period, and the 2025 tariff-driven volatility episode. As Table 4.1 shows, the improvement in every risk-adjusted metric is substantial. The HAC-robust alpha reaches 13.4% per annum ($t = 2.87$, $p = 0.004$), statistically significant at the 1% level, with a downside beta of 0.748 versus 0.869 in-sample, confirming that the strategy becomes more defensive in adverse conditions precisely when this property is most valuable. The information ratio over this window is 1.38. The variance drag reduction (1.28% versus 1.42% for the benchmark) compounds quietly but persistently over multi-year horizons. Tail risk improves as well: Expected Shortfall narrows by 42 basis points relative to the benchmark, and excess kurtosis falls to 0.88, close to the Gaussian benchmark, indicating that the left tail has been substantially reshaped.

Statistical significance of the alpha shift. The increase in alpha from the in-sample to the out-of-sample window is not only economically large ($\Delta\alpha = +10.7\%$ per annum) but reaches conventional statistical significance ($p = 0.044$), while the corresponding change in

beta is negligible and insignificant ($\Delta\beta = -0.021$, $p = 0.73$). A strategy subject to in-sample overfitting would be expected to show the opposite pattern: inflated in-sample alpha degrading out-of-sample. We note that the full-sample alpha (5.6% per annum, $t = 2.46$, $p = 0.014$) confirms this reading at the aggregate level as well. The R^2 of the regression remains stable across the two windows (0.649 pre, 0.636 post), confirming that the structural relationship between strategy and benchmark is unchanged despite the sharp increase in alpha.

Survivorship bias A structural contribution of this work concerns survivorship bias. Rotation strategies applied to individual equities are inherently exposed to this distortion when the eligible universe is defined from the set of companies observable at the time of analysis. By restricting the investable universe to broad exchange-traded funds covering entire asset classes, sectors, and geographies, the strategy avoids this problem entirely: no asset was selected because it is known ex-post to have survived or outperformed.

4.2 Limitations

Several limitations of the proposed framework should be acknowledged. First, financial return series are inherently non-stationary: means, covariances, and higher-order moments evolve over time in ways that cannot be fully captured by any fixed-parameter model. The dual-layer architecture relies on rolling z-scores and fixed scoring weights that were economically motivated rather than estimated, which reduces but does not eliminate sensitivity to regime shifts. The strong out-of-sample performance observed post-2022 is encouraging, yet it cannot be interpreted as a guarantee of future robustness under regimes not represented in the historical sample.

Second, the normalisation of momentum and reversal signals via rolling z-scores implicitly assumes that the cross-sectional score distribution can be approximated by a Gaussian. In practice the tails of financial return distributions are heavier than the normal, and the sample estimates of mean and standard deviation used in the normalisation are themselves noisy, particularly over short rolling windows during turbulent periods.

Third, the turnover analysis presented in this work assumes a fixed one-way transaction cost of five basis points, calibrated to the typical bid-ask spread of liquid ETFs under normal market conditions. In periods of elevated volatility or reduced liquidity, effective transaction costs may rise significantly, eroding the net advantage of active rebalancing relative to a passive benchmark. Although the hysteresis buffer mitigates unnecessary substitution turnover, the residual weight-drift rebalancing cost remains sensitive to the realized volatility environment.

4.3 Future Research

The most immediate extension suggested by this work is the development of a hybrid architecture that combines the complementary strengths of the two models studied here. Rather than applying a deterministic Layer 2 momentum filter followed by a fixed-weight Layer 1 scorer, the selection and weighting steps could both be delegated to learned rankers. A natural implementation would train two distinct LambdaRank models on feature subsets with different statistical profiles: one calibrated on features that exhibit mean-reverting behaviour in the time series dimension, and one on features that carry persistent momentum signals. The outputs of the two rankers could then be combined at the portfolio construction stage, replacing the current deterministic funnel with a data-driven, adaptive allocation layer.

A second direction concerns dynamic weight updating. Rather than rebalancing to a fixed equal-weight or score-proportional target at each period, portfolio (and intra-layer) weights could be updated by an autonomous agent that incorporates real-time macroeconomic scenario probabilities. Recent advances in probabilistic scenario classification, including discriminative language models fine-tuned via reinforcement learning from human feedback (RLHF) and contrastive learning methods (CLM), now make it operationally feasible to maintain an evolving distribution over macro regimes and to condition portfolio tilts on that distribution dynamically. Integrating such a module into the walk-forward execution loop would transform the current fixed-rule overlay into a regime-aware, continuously adapting risk management layer.

Bibliography

Andrew Ang, Robert J Hodrick, Yuhang Xing, and Xiaoyan Zhang. The cross-section of volatility and expected stock returns. *The Journal of Finance*, 61(1):259–299, 2006.

Andrew Ang, Robert J Hodrick, Yuhang Xing, and Xiaoyan Zhang. High idiosyncratic volatility and low unexpected returns: International and further evidence. *Journal of Financial Economics*, 91(1):137–163, 2009.

Clifford S Asness, Andrea Frazzini, and Lasse Heje Pedersen. Leverage aversion and the risk parity puzzle. *Financial Analysts Journal*, 68(1):58–81, 2012.

Clifford S Asness, Tobias J Moskowitz, and Lasse Heje Pedersen. Value and momentum everywhere. *The Journal of Finance*, 68(3):929–985, 2013.

David H Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.

Pedro Barroso and Pedro Santa-Clara. Momentum has its moments. *Journal of Financial Economics*, 116(1):111–120, 2015.

Geert Bekaert, Marie Hoerova, and Marco Lo Duca. Risk, monetary policy, and the vix. *Journal of Monetary Economics*, 60(7):771–783, 2013.

Jean-Philippe Bouchaud and Marc Potters. *Financial Risk and Heavy Tails: Theory and Applications*. Cambridge University Press, 2009.

Chris Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. Learning to rank using gradient descent. In *Proceedings of the 22nd International Conference on Machine Learning*, pages 89–96, 2005.

Christopher J. C. Burges. From ranknet to lambdarank to lambdamart: An overview. Technical Report MSR-TR-2010-82, Microsoft Research, 2010a.

Christopher JC Burges. From ranknet to lambdarank to lambdamart: An overview. *Learning*, 11 (23-581):81, 2010b.

Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Yuzhe Chen, Kexin Zhang, Yibo Wang, and Zhao Dong. Financial time series forecasting with graph neural networks and transformers: A survey. *Expert Systems with Applications*, 215: 119331, 2023.

Kent Daniel and Tobias J Moskowitz. Momentum crashes. *Journal of Financial Economics*, 122(2):221–247, 2016.

Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953, 2009.

Robert F Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

Thomas Fischer and Christopher Krauss. Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669, 2018.

Jerome H Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

Jerome H Friedman. Stochastic gradient boosting. *Computational Statistics & Data Analysis*, 38(4):367–378, 2002.

Richard C Grinold and Ronald N Kahn. *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk*. McGraw-Hill, New York, 2nd edition, 2000.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273, 2020.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.

Rudolf Emil Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, 30:3146–3154, 2017.

Marcos López de Prado. Generalized sharpe ratios, deflated sharpe ratios, and the haircut method. *The Journal of Portfolio Management*, 40(5):125–131, 2014.

Marcos López de Prado. Building diversified portfolios that outperform out of sample. *The Journal of Portfolio Management*, 42(4):59–69, 2016.

Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, Hoboken, NJ, 2018.

Sébastien Maillard, Thierry Roncalli, and Jérôme Teïletche. The properties of equally weighted risk contribution portfolios. *The Journal of Portfolio Management*, 36(4):60–70, 2010.

Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

Richard O Michaud. The markowitz optimization enigma: Is' optimized'optimal? *Financial Analysts Journal*, 45(1):31–42, 1989.

Alan Moreira and Tyler Muir. Volatility-managed portfolios. *The Journal of Finance*, 72(4): 1611–1644, 2017.

Tobias J Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012.

Whitney K Newey and Kenneth D West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Edward Qian. Risk parity portfolios: Efficient portfolios through true risk diversification. *PanAgora Asset Management White Paper*, 2005.

K Geert Rouwenhorst. International momentum strategies. *The Journal of Finance*, 53(1): 267–284, 1998.

J Peter Steidlmayer and Kevin Koy. *Markets and Market Logic*. Porcupine Press, 1984.

Robert E Whaley. Derivatives on market volatility: Hedging tools long overdue. *The Journal of Derivatives*, 1(1):71–84, 1993.